

Beta-Sorted Portfolios*

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Abstract

Beta-sorted portfolios—portfolios comprised of assets with similar covariation with selected risk factors—are a popular tool in empirical finance to analyze models of (conditional) expected returns. Despite their widespread use, little is known of their econometric properties in contrast to comparable procedures such as two-pass regressions. We formally investigate the properties of beta-sorted portfolio returns by casting the procedure as a two-step nonparametric estimator with a nonparametric first step and a beta-adaptive portfolio construction. Our framework rationalizes the well-known estimation algorithm with precise economic and statistical assumptions on the general data-generating process. We provide conditions which ensure valid estimation and inference allowing for a range of hypotheses of interest in financial applications. We show that the rate of convergence of the estimator changes depending on the value of beta. We demonstrate that valid inference depends critically on the object of interest and discuss drawbacks of the widely used Fama-MacBeth variance estimator. To address these limitations, we propose a new variance estimator. We demonstrate the usefulness of our theoretical results in two empirical applications, including one in which we introduce a novel risk factor that captures the business credit cycle and show that it predicts both the cross-sectional and time-series behavior of U.S. stock returns.

Keywords: Beta pricing models, portfolio sorting, nonparametric estimation, partitioning, kernel regression, smoothly-varying coefficients, Fama-MacBeth variance estimator.

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1 Introduction

Deconstructing expected returns into asset-specific factor loadings and corresponding prices of risk for interpretable factors is an evergreen pursuit in the empirical finance literature. When factors are observable, there are two workhorse approaches that continue to enjoy widespread use. The first approach, Fama-MacBeth two-pass regressions, has been extensively studied in the financial econometrics literature.¹ The second approach, which we refer to as beta-sorted portfolios, has received scant attention in the econometrics literature despite its empirical popularity.²

The implementation of beta-sorted portfolios entails the following two-step procedure, which incorporates a beta-adaptive portfolio construction (see, e.g., [Bali et al. 2016](#)). In a first step, time-varying risk factor exposures (betas) are estimated through (backward-looking) weighted time-series regressions of asset returns on the observed factors. The most popular implementation uses rolling window regressions, often with a choice of a five-year window. In a second step, the estimated factor exposures, based on data up to the previous period, are ordered and used to group assets into portfolios. These portfolios then represent assets with a similar exposure to the risk factors, and the magnitude of the return differential for differently exposed assets is used to assess the compensation for bearing this common risk. Most frequently this is achieved by differencing the portfolio returns from the two most extreme portfolios. Finally, an average over time of these return differentials is taken to infer whether the risk is priced unconditionally—whether the long/short portfolio earns systematic (and significant) excess returns. Notwithstanding the simple and intuitive nature of the methodology, little is known of the formal properties of this estimator and its associated inference procedures.

We provide a comprehensive framework to study the economic and statistical properties of beta-sorted portfolios. We first translate the two-step estimation algorithm with beta-adaptive portfolio construction into a corresponding econometric model. We show that the model has key features which are important to consider for valid interpretation of the empirical results. Notably, we do not impose no-arbitrage conditions. Instead, the restrictions implied by no arbitrage become testable

¹See, for example, [Jagannathan and Wang \(1998\)](#), [Shanken and Zhou \(2007\)](#), [Kleibergen \(2009\)](#), [Ang et al. \(2020\)](#), [Gospodinov et al. \(2014\)](#), [Adrian et al. \(2015\)](#), [Gagliardini et al. \(2016\)](#), [Raponi et al. \(2020\)](#), [Giglio and Xiu \(2021\)](#) and many others. For a recent survey, see [Gagliardini et al. \(2020\)](#).

²The empirical literature using beta-sorted portfolios is extensive. For a textbook treatment, see [Bali et al. \(2016\)](#), and for a few recent papers see, for example, [Boons et al. \(2020\)](#), [Chen et al. \(2021\)](#), [Eisdorfer et al. \(2022\)](#), [Goldberg and Nozawa \(2021\)](#), [Fan et al. \(2022\)](#), [Chen et al. \(2023\)](#), and [Chen et al. \(2025\)](#).

hypotheses. Furthermore, our framework precisely clarifies how the dynamics of the risk factors relate to the functional form of conditional expected returns. Within this framework, we introduce general sampling assumptions allowing for time-varying factor loadings, serially-correlated factors, and conditional heteroskedasticity across time and assets. We then study the asymptotic properties of the beta-sorted portfolio estimator and associated inference procedures in settings with large cross-sectional, n , and time-series, T , sample sizes (i.e., $n, T \rightarrow \infty$).

Our analysis yields new methodological and theoretical results for this popular procedure. At any fixed date, the beta-sorted portfolio estimator is centered on the sum of two terms: conditional expected returns and a linear combination of the common risk-factor innovations. By a law of large numbers, the time average of the second term converges to zero, so the estimator targets the sample average of conditional expected returns (SACER), which we argue is the natural object of interest. We establish uniform consistency for this time-averaged estimator under transparent high-level conditions. The key requirements are that the first-step beta estimates become uniformly accurate, the portfolios become sufficiently homogeneous in beta values, and each portfolio contains enough assets. Importantly, this consistency result does not impose a particular dynamic model for the betas and can accommodate commonly used parametric and nonparametric specifications (e.g., [Adrian et al. 2015](#), [Gagliardini et al. 2016](#), [Gu et al. 2020](#)).

These findings have direct implications for how beta-sorted portfolios should be constructed and reported. To estimate conditional expected returns for a fixed value of β , a researcher must track the portfolio containing that value in each period. Mechanically averaging portfolios with the same rank over time, as is commonly done, can combine different beta ranges as the cross-sectional distribution of these risk-factor exposures generally changes. The first-step window size, H , and the number of portfolios, J in the single-factor case, should both adapt to the available time-series and cross-sectional sample sizes rather than be fixed at conventional choices such as five years and ten portfolios. We develop data-driven choices for H and J and provide practical guidance for their use in estimation and inference.

Our distribution theory uncovers a sharp difference in the estimator’s behavior at nonzero and at zero risk-factor exposures. For any fixed nonzero evaluation point, the estimator retains a common risk-factor component that is unaffected by the number of assets in each cross-section, so that the time-averaged estimator converges at only a $\sqrt{T}$ rate. When evaluated at the origin, this

common component disappears and idiosyncratic variation—including the additional uncertainty from estimating the risk-factor exposures—becomes the dominant source of uncertainty. Under our stated conditions, the estimator can attain the faster $\sqrt{nT/J^d}$ rate. This special role of zero follows from the model’s factor structure and extends beyond the beta-sorted portfolio estimator.

We propose a variance estimator that separately estimates the common risk-factor and idiosyncratic components and accounts for the dependence created by overlapping first-step windows. We show that it provides valid, although potentially conservative, inference. To address the changing behavior near zero, we develop a new inference procedure that remains valid across different values of β , including at and near the origin. The procedure uses a data-driven rule to identify the region near zero and a conservative variance adjustment to ensure that the associated confidence intervals attain at least their nominal coverage asymptotically. We also show that the ubiquitous Fama-MacBeth (FM) variance estimator ([Fama and MacBeth, 1973](#)) is generally conservative for inference on the SACER at nonzero evaluation points but may not be valid at the origin, where it can either overestimate or underestimate uncertainty.

Our framework accommodates multiple risk factors and clarifies the trade-offs among common ways of constructing portfolios. Independent sorts form J groups based on each of the d estimated risk-factor exposures and then intersect them, producing J^d portfolios that may contain too few assets as d increases. Estimating several betas but sorting on only one avoids this issue. However, this approach changes the object being estimated. We characterize this alternative estimand and explain how our framework can be adapted to conduct inference on it. Conditional sorts first group assets by one estimated risk-factor exposure and then divide each group further using additional exposures. This approach can also help avoid empty portfolios, but does not eliminate the dimensionality problem. In this case, recovering the SACER at a fixed beta requires tracking the exposure ranges represented by each portfolio. Averaging returns or spreads by sequential rank across groups, as is commonly done in practice, generally combines different exposure regions, complicating interpretation.

Certain linear combinations of beta-sorted portfolio returns eliminate the common risk-factor component and, when constructed appropriately, can attain the same faster rate. A butterfly portfolio is one such “factor-cancelling contrast” and measures a discrete second difference in the SACER, thereby capturing curvature in the relationship between expected returns and risk-factor

exposures. Separately, we develop a test of the affine restriction implied by no arbitrage for raw returns and the corresponding linear restriction for excess returns.

From a theoretical perspective, beta-sorted portfolios present a number of technical challenges since the estimator relies on two nested nonparametric estimation steps together with a portfolio construction based on a first-step nonparametric generated regressor. More precisely, risk-factor exposures estimated nonparametrically in the first step enter directly into the nonsmooth partitioning scheme which complicates the analysis.³ To our knowledge, we are the first to prove validity of such an approach.

This paper is most related to the large literature studying asset pricing models with observable factors.⁴ Given our focus on conditional asset pricing models with large panels in both the cross-sectional and time-series dimensions, this paper is most closely related to [Gagliardini et al. \(2016\)](#) (see also [Gagliardini et al., 2020](#)). The authors introduce a general framework and econometric methodology for inference in large-dimensional conditional factor models under no-arbitrage restrictions. They model risk-factor exposures as parametric functions of observable variables and establish conditions for consistent estimation and inference on the prices of risk. Although the statistical model under study shares important similarities with the setup of [Gagliardini et al. \(2016\)](#), there are substantial differences, and the models explored previously in the literature do not nest our setup. For example, our framework for the classical beta-sorted portfolio estimator accommodates data-generating processes that do not necessarily exclude arbitrage opportunities and allow risk-factor exposures to vary smoothly over time. Furthermore, unlike the existing literature (e.g., [Adrian et al. 2015](#), [Gagliardini et al. 2016](#)), we establish valid estimation and inference without specifying the functional form of the conditional mean of the risk factors.

Our paper is also related to the financial econometrics literature on nonparametric estimation and inference. In particular, the two steps of the beta-sorted portfolio algorithm align individually with [Ang and Kristensen \(2012\)](#), who study kernel regression estimators of time-varying alphas and

³For analysis of partitioning-based nonparametric estimators see [Cattaneo et al. \(2020b\)](#) and references therein. Partitioning-based estimators with random basis functions have been recently studied in [Cattaneo et al. \(2020a\)](#) and [Cattaneo et al. \(2024\)](#), but in those papers the conditioning variables are observed, while here the conditioning variable is generated using a preliminary time-series (nonparametric) regression, and therefore prior results are not applicable to the settings considered herein.

⁴See, for example, [Goyal \(2012\)](#), [Nagel \(2013\)](#), [Gospodinov and Robotti \(2013\)](#), or [Gagliardini et al. \(2020\)](#) for surveys. A related literature endeavors to jointly estimate factor loadings and *latent* risk factors. See, for example, [Connor and Linton \(2007\)](#), [Connor et al. \(2012\)](#), [Kelly et al. \(2019\)](#), and [Borri et al. \(2024\)](#).

betas, and [Cattaneo et al. \(2020a\)](#), who study portfolio sorting estimators given observed individual characteristic variables. However, the individual results from each of these papers cannot be applied in our multi-step setting. Furthermore, the linkage between the two steps, through the role of the generated (nonparametrically estimated) regressor in the second-stage nonparametric partitioning estimator, represents a substantial technical challenge and has not been studied before.

We demonstrate the practical usefulness of our estimation and inference results in two empirical applications. First, we introduce a new risk factor—a measure of the business credit cycle—based on the Federal Reserve’s Senior Loan Officer Opinion Survey and the ISM Manufacturing Index. We show that this factor is predictive of both the cross-sectional and time-series behavior of U.S. stock returns. Second, we revisit the capital share factor from [Lettau et al. \(2019\)](#). Using our new inference procedures, we find no evidence that the capital share risk factor is priced in the cross-section of U.S. stock returns.

We collect additional information and technical details in two Supplemental Appendices. The Supplemental Appendix: Theory (SA-Theory) develops the technical foundations for our results, including primitive conditions, additional theoretical results, and proofs. The Supplemental Appendix: Applied (SA-Applied) provides implementation details, data-driven guidance for tuning-parameter selection, simulation evidence, and the second empirical application. Unless otherwise stated, asymptotic limits are taken as $n, T \rightarrow \infty$, with H and J varying as specified below. We use $O_{\mathbb{P}}(1)$ and $o_{\mathbb{P}}(1)$ for sequences that are bounded in probability and converge to zero in probability, respectively. For positive random sequences a_n and b_n , $a_n \asymp_{\mathbb{P}} b_n$ means that both a_n/b_n and b_n/a_n are $O_{\mathbb{P}}(1)$. For positive deterministic sequences $a_n \asymp b_n$ means both a_n/b_n and b_n/a_n are $O(1)$. Finally, for a set $S \subseteq \mathbb{R}^d$, define its diameter as $\text{diam}(S) = \sup\{\|s - s'\| : s, s' \in S\}$.

2 Model setup

We introduce a general econometric model of asset returns and show how it aligns with the two steps that comprise the beta-sorted portfolio algorithm. We explain the rationale behind the form of the estimator and discuss relevant properties of the model.

2.1 Modeling returns

Let R_{it} denote the return of asset i at time t , and f_t a vector of observable risk factors with $f_t \in \mathbb{R}^d$.

We assume that asset returns are generated by the linear stochastic coefficient model,

$$R_{it} = \alpha_{it} + \beta_{it}^\top f_t + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} | \mathcal{F}_{t-1}, f_t] = 0, \quad i = 1, \dots, n_t, \quad t = 1, \dots, T, \quad (2.1)$$

where α_{it} and β_{it} are random coefficients, ε_{it} is an idiosyncratic error term, and $\mathcal{F}_t$ is the information set up to time t .⁵ The sigma field $\mathcal{F}_t$ will, in general, depend on T and $\{n_s : s < t\}$ but we suppress this dependence for simplicity of notation. We assume that α_{it} and β_{it} are both $\mathcal{F}_{t-1}$ -measurable so we have that $\mathbb{E}[R_{it} | \mathcal{F}_{t-1}, f_t] = \alpha_{it} + \beta_{it}^\top f_t$. Finally, notice that equation (2.1) accommodates an unbalanced panel. Although each n_t may be different, we assume that they all grow at the same rate which ensures that each cross-section contributes to the asymptotic properties of the estimator (i.e., $n_t \asymp n$ for $t = 1, \dots, T$).

To obtain the structural form of our model, we assume that there exists a non-random function $\mu_0(\cdot)$ such that

$$\mathbb{E}(R_{it} | \mathcal{F}_{t-1}) = \mu_0(\beta_{it}; \mathcal{G}_{t-1}) = \mu_t(\beta_{it}), \quad (2.2)$$

where $\mathcal{G}_t$ is the sub-sigma field of $\mathcal{F}_t$ generated only by i -invariant random variables such as time-series like f_t . Equation (2.2) restricts conditional expected returns to be a function of β_{it} only, but it allows the functional form to be random and to vary with past realizations of economy-wide random variables.⁶ This restriction requires assets with the same risk-factor exposures (as measured by β_{it}) to have the same conditional expected return in a given period. For the discussion in this subsection, we take R_{it} to denote excess returns for simplicity of presentation. If R_{it} instead denotes raw returns, $\mu_t(0)$ may be interpreted as the zero-beta rate at time t .

Since the economy's state variables will reside in $\mathcal{G}_t$, a natural restriction is to impose that

$$\mathbb{E}[f_t | \mathcal{F}_{t-1}] = \mathbb{E}[f_t | \mathcal{G}_{t-1}], \quad t = 1, 2, \dots, T. \quad (2.3)$$

⁵For an alternative example of a random coefficient model tailored to a financial application, see [Barras et al. \(2022\)](#).

⁶This allows considerable flexibility in conditional expected returns, including time variation in the importance of economy-wide variables and in the strength of return predictability (e.g., [Farmer et al., 2023](#)).

This economic restriction also ensures that the systematic component of R_{it} is determined only by β_{it} and economy-wide variables. Using equation (2.3) along with equations (2.1) and (2.2), we obtain,

$$\alpha_{it} = \alpha_t(\beta_{it}), \quad \alpha_t(\beta) = \mu_t(\beta) - \beta^\top \mathbb{E}(f_t | \mathcal{G}_{t-1}), \quad (2.4)$$

which clarifies the restriction on α_{it} implied by equation (2.2). Equation (2.4) is the nonparametric analog to a reduced-rank restriction in parametric models as it imposes cross-equation restrictions between the linear coefficients in equation (2.1). In fact, equation (2.4) nests as a special case the assumption of no arbitrage which directly implies a parametric reduced-rank regression model (e.g., [Adrian et al., 2015](#)). As shown in [Gagliardini et al. \(2016\)](#), in the absence of arbitrage opportunities, there exists a λ_t such that $\mu_t(\beta) = \beta^\top \lambda_t$ for all β which implies $\alpha_t(\beta) = \beta^\top (\lambda_t - \mathbb{E}(f_t | \mathcal{G}_{t-1}))$.

Finally, combining equations (2.1) and (2.4), we arrive at the structural form

$$R_{it} = \mu_t(\beta_{it}) + \beta_{it}^\top (f_t - \mathbb{E}[f_t | \mathcal{G}_{t-1}]) + \varepsilon_{it}. \quad (2.5)$$

Equation (2.5) may be compared to the standard beta pricing model (e.g., [Cochrane, 2005](#), Chapter 12) and generalizations thereof (e.g., [Cochrane, 1996](#); [Adrian et al., 2015](#); [Gagliardini et al., 2016](#)). The most noteworthy difference is the presence of the (possibly) nonlinear, time-varying function $\mu_t(\beta_{it})$ in equation (2.5). When $\mu_t(\cdot)$ is nonlinear, the model departs from the no-arbitrage pricing restriction discussed above. Such departures can be motivated by structural models featuring violations of the law of one price. For example, equation (2.5) could arise in the margin-constraints model of [Gârleanu and Pedersen \(2011\)](#) under the assumption that the security's margin is a nonlinear function of its (past) beta.

2.2 Relation to the two-step estimator

To understand the rationale behind the beta-sorted portfolios algorithm, it is instructive to (temporarily) make the simplifying assumption that β_{it} is observable. Let us define systematic realized returns as

$$M_t(\beta) = \mu_t(\beta) + \beta^\top (f_t - \mathbb{E}[f_t | \mathcal{G}_{t-1}]), \quad (2.6)$$

which implies that $M_t(\beta_{it}) = \alpha_{it} + \beta_{it}^\top f_t$ and

$$R_{it} = M_t(\beta_{it}) + \varepsilon_{it}. \quad (2.7)$$

This equation makes clear that, for a fixed time period, we can only consistently (nonparametrically as $n_t \rightarrow \infty$) estimate the unknown function $M_t(\cdot)$ rather than the direct object of interest $\mu_t(\cdot)$. This comes about because both terms in $M_t(\beta)$ are an unknown transformation of β as they involve $\mu_t(\cdot)$ and $\mathbb{E}[f_t|\mathcal{G}_{t-1}]$ neither of which are observed. However, its sample average may be written as,

$$\frac{1}{T} \sum_{t=1}^T M_t(\beta) = \frac{1}{T} \sum_{t=1}^T \mu_t(\beta) + \frac{1}{T} \sum_{t=1}^T \beta^\top (f_t - \mathbb{E}[f_t|\mathcal{G}_{t-1}]). \quad (2.8)$$

Importantly, even though we do not observe $\mathbb{E}[f_t|\mathcal{G}_{t-1}]$, the second term has summands, $\beta^\top (f_t - \mathbb{E}[f_t|\mathcal{G}_{t-1}])$, which are a martingale difference sequence with respect to $\mathcal{G}_t$. Under mild conditions a law of large numbers will hold, and this sample average will converge to zero in probability so that

$$\frac{1}{T} \sum_{t=1}^T M_t(\beta) = \frac{1}{T} \sum_{t=1}^T \mu_t(\beta) + o_{\mathbb{P}}(1). \quad (2.9)$$

This demonstrates how $T^{-1} \sum_{t=1}^T \mu_t(\beta)$, the sample average of conditional expected returns, can be consistently estimable even if the individual $\mu_t(\beta)$ are not. A notable exception occurs when $\beta = 0$ as the linear term in $M_t(\beta)$ is eliminated (see Remark 2.2 for further discussion).

In practice, we must estimate $M_t(\cdot)$ from the data (which requires estimates of the β_{it}) to obtain a feasible estimator of $T^{-1} \sum_{t=1}^T \mu_t(\beta)$. In this setting the choice of estimator is informed by additional considerations beyond just statistical properties. Specifically, empirical practice in finance has revolved around two considerations. First, the estimator should have the interpretation of a feasible investment strategy, so that estimates of expected returns and their statistical significance can be directly assessed in economic terms. Second, the estimator should have a “real-time” formulation, i.e., when an additional time period of data is added, prior estimates are unaffected.

Taking these two considerations in mind translates to the following two-step procedure which is ubiquitous in the empirical finance literature:

Step One: Using only information up to time $t - 1$, construct $\hat{\beta}_{it}$ for $i = 1, \dots, n_t$. The classical implementation uses rolling window OLS regressions,

$$\begin{pmatrix} \hat{\alpha}_{it} \\ \hat{\beta}_{it} \end{pmatrix} = \arg \min_{a \in \mathbb{R}, b \in \mathbb{R}^d} \sum_{s=1}^H (R_{i(t-s)} - a - b^\top f_{t-s})^2, \quad (2.10)$$

where H is the window size. This approach can be formally justified under the assumption that α_{it} and β_{it} are smoothly varying (random) coefficients which we discuss further in Section 4.

Step Two: Based on step one, group assets together with similar values of $\hat{\beta}_{it}$ and form portfolios. The classical implementation uses marginal quantiles from the cross-sectional distribution of $\hat{\beta}_{it}$ to separate the assets. Next, using the individual returns for time t , calculate the return for each portfolio. Finally, average the portfolio returns over time to assess the unconditional risk properties of the investment strategy.

Portfolios have been established as the fundamental investment object since at least [Markowitz \(1952, 1959\)](#) and [Tobin \(1958\)](#) and so are the natural choice to link returns, R_{it} , to risk exposures, β_{it} . However, since the β_{it} are unobserved they must first be estimated in a preliminary step. Moreover, to ensure the portfolio returns could have been earned in real time (without look ahead bias) each β_{it} is estimated using only information up to time $t - 1$.

To formally write the beta-sorted portfolio estimator, let $\hat{\Delta}_t = \{\hat{P}_{jt} : 1 \leq j \leq \bar{J}_t\}$ be a collection of nonoverlapping cells covering the support $\mathcal{B}$ and all estimated betas, based on $\{\hat{\beta}_{it} : i = 1, \dots, n_t\}$. The $\hat{P}_{jt}$ are portfolio designations; specifically, each asset i is allocated to a portfolio, $\hat{P}_{jt}$, depending on its value for $\hat{\beta}_{it}$. For example, in the classical implementation, portfolio designation is governed by the marginal quantiles of the (cross-sectional) distribution of $\hat{\beta}_{it}$, i.e., when $d = 1$,

$$\begin{aligned} \hat{P}_{1t} &= \left[\min_{1 \leq i \leq n_t} \hat{\beta}_{it}, \hat{F}_{\hat{\beta},t}^{-1}(1/\bar{J}_t) \right), \\ \hat{P}_{jt} &= \left[\hat{F}_{\hat{\beta},t}^{-1}((j-1)/\bar{J}_t), \hat{F}_{\hat{\beta},t}^{-1}(j/\bar{J}_t) \right), \quad j = 2, \dots, \bar{J}_t - 1, \\ \hat{P}_{\bar{J}_t} &= \left[\hat{F}_{\hat{\beta},t}^{-1}((\bar{J}_t - 1)/\bar{J}_t), \max_{1 \leq i \leq n_t} \hat{\beta}_{it} \right], \end{aligned} \quad (2.11)$$

where $\hat{F}_{\hat{\beta},t}^{-1}(\cdot)$ is the empirical quantile function. In practice, the first and last cells are extended

outward, as needed, so that the partition covers the full support of the betas. When $d > 1$, the classical implementation uses the tensor (Cartesian) product of the partition in equation (2.11) across the d dimensions. It is important to emphasize that the portfolio that any specific β resides in may change over time. For example, when $d = 1$, it may be that assets with values of β near $1/2$ fall in the sixth portfolio at times and the fifth portfolio at other times and so on.

To formalize the beta-sorted portfolio estimator we need to keep track of which portfolio any specific value of β resides in. We do this via

$$\hat{p}_t(\beta) = \left(\mathbb{1}(\beta \in \hat{P}_{1t}), \dots, \mathbb{1}(\beta \in \hat{P}_{J_t t}) \right)^\top. \quad (2.12)$$

The vector, $\hat{p}_t(\beta)$, is comprised of a single element equal to one and all other elements equal to zero. For example, if the value of $\hat{\beta}_{it}$ resides in the first portfolio, $\hat{P}_{1t}$, then $\hat{p}_t(\hat{\beta}_{it}) = (1, 0, \dots, 0)^\top$, and the return for the first portfolio at time t is

$$\frac{1}{\hat{n}_{1t}} \sum_{i=1}^{n_t} \mathbb{1}\{\hat{\beta}_{it} \in \hat{P}_{1t}\} R_{it}, \quad \hat{n}_{1t} = \sum_{i=1}^{n_t} \mathbb{1}\{\hat{\beta}_{it} \in \hat{P}_{1t}\}. \quad (2.13)$$

More generally, the beta-sorted portfolio estimator at time t is

$$\widehat{M}_t(\beta) = \hat{p}_t(\beta)^\top \left(\sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \hat{p}_t(\hat{\beta}_{it})^\top \right)^{-1} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) R_{it} \quad (2.14)$$

and the corresponding sample average estimator is

$$\hat{\mu}(\beta) = \frac{1}{T-H} \sum_{t=H+1}^T \widehat{M}_t(\beta). \quad (2.15)$$

The estimator is comprised of $T - H$ individual estimates, $\widehat{M}_t(\beta)$, motivated by equation (2.8). Notice that $\widehat{M}_t(\beta)$ can be interpreted as the fitted value from a cross-sectional OLS regression of R_{it} on $\hat{p}_t(\hat{\beta}_{it})$ and evaluated at $\hat{p}_t(\beta)$. Furthermore, it is straightforward to show that if $\beta \in \hat{P}_{jt}$, then $\widehat{M}_t(\beta)$ may be equivalently written as,

$$\widehat{M}_t(\beta) = \frac{1}{\hat{n}_{jt}} \sum_{i=1}^{n_t} \mathbb{1}\{\hat{\beta}_{it} \in \hat{P}_{jt}\} R_{it}, \quad \hat{n}_{jt} = \sum_{i=1}^{n_t} \mathbb{1}\{\hat{\beta}_{it} \in \hat{P}_{jt}\}. \quad (2.16)$$

The sample average in the definition of $\hat{\mu}(\beta)$ begins at time $H + 1$ to reflect the need for an initial sample period (of size H) to enable estimation of $\beta_{i(H+1)}$.⁷

We will provide conditions under which $\hat{\mu}(\beta)$, given by equation (2.15), is uniformly consistent for the sample average of conditional expected returns (SACER) defined as

$$\bar{\mu}_T(\beta; H) = \frac{1}{T - H} \sum_{t=H+1}^T \mu_t(\beta). \quad (2.17)$$

The estimator, $\hat{\mu}(\beta)$, thus provides an estimate of the SACER for a generic asset with risk exposures given by the evaluation point, β .

Remark 2.1 (Comparison to the typical implementation approach). There is one subtle but important difference between equation (2.15) and the typical implementation of beta-sorted portfolios.⁸ In existing practice, $\bar{J}_t = \bar{J}$ for all t , and the reported $\hat{\mu}(\cdot)$ takes on only $\bar{J}$ values as the specific evaluation point β is not considered. For example, the first element of the reported $\hat{\mu}(\cdot)$ is the average return across time associated with the first portfolio, $\hat{P}_{1t}$. However, if the cross-sectional distribution of the β_{it} is changing over time then there is no guarantee that a chosen portfolio (e.g., the fifth portfolio) always represents assets with sufficiently similar betas.⁹ Therefore, the typical implementation does not generally estimate the SACER at a fixed value of β ; rather, it averages returns over beta regions that may change over time. This distinction is particularly important when we are interested in expected returns at intermediate values of β or in tests of monotonicity or other shape restrictions.

Remark 2.2 (Evaluation at $\beta = 0$). As discussed above, the case where $\beta = 0$ is distinctive in this model. Observe that

$$\mathbb{E}[R_{it} | \mathcal{G}_{t-1}, \beta_{it} = 0] = M_t(0) = \mu_t(0), \quad (2.18)$$

so we no longer require the time averaging described by equation (2.8) for $\mu_t(0)$ to be consistently estimable. The disappearance of the common risk-factor component at zero is a property of the

⁷We use H as the length of the initial sample for convenience of notation. A longer burn-in period could also be used.

⁸We distinguish between “classical” beta-sorted portfolios and the “typical” implementation. By the former we mean the use of rolling window regressions to estimate β_{it} and marginal quantiles of the distribution of $\hat{\beta}_{it}$ to form portfolios. By the latter we mean choosing a fixed number of portfolios (often ten) and reporting only the average returns of those portfolios.

⁹In fact, there may be no value of β that resides in the same ordered portfolio across all time periods.

data-generating process and is not specific to the choice of nonparametric estimator. This distinction will be made throughout the presentation of the theoretical results in the subsequent sections. However, it is important to emphasize that primary interest is generally in values of β which are of larger magnitude.

3 Consistency of beta-sorted portfolios

We first prove consistency of the beta-sorted portfolio estimator under minimal conditions. It is important to emphasize the challenging structure of the estimator as the (generally nonparametrically) generated regressors, $\hat{\beta}_{it}$, enter the second stage through non-smooth indicator functions, $\hat{p}_t(\hat{\beta}_{it})$. Further, the first step and the second step share the same underlying data used to estimate β_{it} and managing this dependence presents a number of technical challenges.

We first provide conditions on the DGP.

Assumption 1 (DGP). *The data satisfy equations (2.1)–(2.3) and (i) β_{it} has the compact support $\mathcal{B}$ for each $i = 1, \dots, n_t$ and $t = 1, \dots, T$; (ii) there exists a finite constant $C > 0$ such that $|\mu_t(\beta) - \mu_t(\beta')| \leq C\|\beta - \beta'\|$ for all $\beta, \beta' \in \mathcal{B}$ and $H + 1 \leq t \leq T$; (iii) ε_{it} is independent over i for each t conditional on $\mathcal{F}_{t-1}$; (iv) $\max_{1 \leq t \leq T} \max_{1 \leq i \leq n_t} \mathbb{E}[|\varepsilon_{it}|^q | \mathcal{F}_{t-1}]$ and $\max_{1 \leq t \leq T} \mathbb{E}[\|\tilde{f}_t\|^q | \mathcal{F}_{t-1}]$ are bounded for some $q > 2$ where $\tilde{f}_t = f_t - \mathbb{E}[f_t | \mathcal{G}_{t-1}]$.*

Assumption 1(i)–(ii) are standard assumptions in the nonparametric literature which rule out discontinuities and other pathologies that would invalidate standard nonparametric estimation approaches. Assumption 1(iii) restricts the cross-sectional dependence properties of ε_{it} . This type of sampling assumption was introduced in Andrews (2005) and has been utilized in the financial econometrics literature by Gagliardini et al. (2016) and Cattaneo et al. (2020a). The assumption is quite general and allows for higher-order dependence in the distribution of ε_{it} ; for example, it accommodates conditional heteroskedasticity both cross sectionally and over time. In Remark 4.3 of Section 4, we discuss weakening this assumption. Finally, Assumption 1(iv) comprises standard moment conditions on the factors and idiosyncratic errors.

Since the beta-sorted portfolio estimator consists of two steps, we must impose some conditions on the first-step estimator. However, we only impose minimal assumptions on the estimators of β_{it} .

Assumption 2 (First step estimator). $\widehat{\beta}_{it}$ is $\mathcal{F}_{t-1}$ -measurable, $H \leq cT$ for a constant $c \in (0, 1)$, and there exists some sequence of positive constants R_{FS} such that

$$\max_{H+1 \leq t \leq T} \max_{1 \leq i \leq n_t} \|\widehat{\beta}_{it} - \beta_{it}\| = O_{\mathbb{P}}(R_{\text{FS}}).$$

Assumption 2 allows for an arbitrary collection of first-step estimators. It requires each $\widehat{\beta}_{it}$ to only use information available before R_{it} is realized and defines their uniform convergence rate, R_{FS} . We will impose that $R_{\text{FS}} = o(1)$ which is a natural (and generally necessary) assumption when working with generated regressors. For the rolling window OLS estimator in equation (2.10), Lemma SA-II.2.3 of SA-Theory provides primitive conditions under which

$$R_{\text{FS}} = \sqrt{\frac{\log(nT)}{H}} + \frac{(nT)^{1/q} \log(nT)}{H} + B_{\text{FS}}, \quad (3.1)$$

where the first two terms capture the variability of the estimator and B_{FS} captures the bias from approximating the time-varying coefficients by a constant function in the window. Thus, we require H to diverge with T in such a way that ensures that all of these terms vanish asymptotically.

Importantly, we only require that uniformly consistent estimators of the β_{it} are available but they are not required to be of the form given by equation (2.10). Assumption 2 covers, for example, cases where a parametric or nonparametric form for the β_{it} is specified in terms of additional variables (e.g. Connor et al., 2012; Gagliardini et al., 2016; Gu et al., 2020) so long as suitable estimators are available.

We now introduce the following assumption which ensures that the probability of an empty portfolio vanishes as the sample size grows.

Assumption 3 (Partition). Let $\widehat{n}_{\min} = \min_{H+1 \leq t \leq T} \min_{1 \leq j \leq \bar{J}_t} \widehat{n}_{jt}$. There exists some positive sequence $n_{\min} > 0$ such that $\widehat{n}_{\min} \geq n_{\min}$ with probability approaching one.

Our first main result is the following:

Theorem 3.1. Suppose Assumptions 1–3 hold and the following conditions are satisfied:

- (a) $R_{\text{FS}} = o(1)$,
- (b) $\widehat{R}_{\text{D}} = o_{\mathbb{P}}(1)$ where $\widehat{R}_{\text{D}} = \max_{H+1 \leq t \leq T} \max_{1 \leq j \leq \bar{J}_t} \text{diam}(\widehat{P}_{jt})$,

- (c) $\frac{\log(nT)}{n_{\min}} = o(1)$,
- (d) $\limsup_{n,T \rightarrow \infty} \frac{(nT)^{2/q} \log(nT)}{n_{\min}} \leq C$ for some constant C .

Then,

$$\sup_{\beta \in \mathcal{B}} \left| \hat{\mu}(\beta) - \bar{\mu}_T(\beta; H) \right| = o_{\mathbb{P}}(1).$$

Theorem 3.1 shows that the beta-sorted portfolio estimator is uniformly consistent under general conditions that follow naturally from the structure of the estimator: Condition (a) requires that the first-step estimators of the β_{it} approximate their targets sufficiently well. Combining this requirement with condition (b) ensures that assets within a portfolio have increasingly similar true betas. Condition (c) ensures that all portfolios are comprised of a sufficient number of assets. Together, conditions (c) and (d) ensure that the contribution of the idiosyncratic errors vanishes asymptotically.

If the partition is “quasi-uniform” (so that no portfolio covers an arbitrarily large or small portion of the support) and uses a number of intervals of order J along each of the d coordinates, and each portfolio contains roughly the same number of assets, then it can be shown that

$$\max_{H+1 \leq t \leq T} \max_{1 \leq j \leq \bar{J}_t} \text{diam}(\hat{P}_{jt}) = O_{\mathbb{P}}(J^{-1}), \quad n_{\min} \asymp \frac{n}{J^d}, \quad (3.2)$$

where $\bar{J}_t \asymp J^d$ uniformly in t . In this case, conditions (b)–(d) require that $J \rightarrow \infty$, $J^d \log(nT)/n \rightarrow 0$, and $J^d (nT)^{2/q} \log(nT)/n = O(1)$. Consequently, J performs a similar balancing act as H ; specifically, J must grow fast enough to eliminate the approximation bias from grouping assets with different betas together, but slowly enough to ensure that each portfolio contains a sufficient number of assets. For the marginal-quantile portfolio construction introduced in equation (2.11), Lemma SA-II.3.3 of SA-Theory provides primitive conditions under which the resulting partition is quasi-uniform over appropriately defined interior portfolios.

Apart from assuming a compact support and the availability of uniformly consistent first-step estimators, consistency of $\hat{\mu}(\cdot)$ requires no direct restrictions on the time-series or cross-sectional properties of β_{it} . For example, the β_{it} can be random, can be correlated in the cross-section and the time-series, and could depend on additional variables in a (possibly) nonlinear fashion. Furthermore, the factors, f_t , are only required to satisfy the (conditional) moment restrictions given

by Assumption 1(iv), but otherwise have no restrictions on their serial dependence properties.

Remark 3.2 (Technical details on Theorem 3.1). Under the conditions of Theorem 3.1, Theorem SA-I.2.1 of SA-Theory establishes the stronger result that, for any arbitrarily slowly diverging sequence ℓ_n (e.g., $\ell_n = \log \log(nT)$), there exists a fixed, finite constant $C > 0$ such that, with probability approaching one, simultaneously for all $\beta \in \mathcal{B}$,

$$|\widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H)| \leq C\ell_n \left[R_{\text{FS}} + \widehat{R}_{\text{D}} + \sqrt{\frac{\log(nT)}{n_{\min}}} + \frac{\|\beta\|}{\sqrt{T}} \right]. \quad (3.3)$$

Unlike the uniform consistency result given in Theorem 3.1, this bound identifies the components that can determine the accuracy of the estimator. It is noteworthy that the result does not require a specific partitioning rule (i.e., portfolio construction) since only the maximum diameter and the minimum number of assets in any one portfolio enter the bound. The final term arises from the unforecastable component of the risk factors and is proportional to the norm of the evaluation point β . For a fixed nonzero β , this term is unaffected by the cross-sectional sample size, whereas it vanishes at $\beta = 0$. The disappearance of this term at $\beta = 0$ is the source of the differential convergence rates of the estimator that we establish in Section 4.

3.1 Object of interest: SACER vs. PACER

In Theorem 3.1 we showed that $\widehat{\mu}(\beta)$ is a consistent estimator of the SACER. Following the arguments in Section 2, the SACER is the natural candidate for our object of interest. However, it is important to point out that the SACER is a sample average of (generally) random quantities and, hence, random itself. The randomness arises from the presence of $\mu_t(\cdot)$ which depends on the i -invariant sigma field $\mathcal{G}_{t-1}$. We can contrast SACER to its population counterpart, the population average of conditional expected returns (PACER). To do so we need the following assumption:

Assumption P (PACER). *There exists a non-random function $\mu(\cdot)$ such that $\sup_{\beta \in \mathcal{B}} |\bar{\mu}_T(\beta; H) - \mu(\beta)| = o_{\mathbb{P}}(1)$.*

Assumption P imposes a uniform convergence condition on the SACER and defines the PACER as its non-random (probability) limit. In the special case where $\mu_t(\cdot) = \mu(\cdot) \forall t$, the SACER and PACER are equal.

The SACER and the PACER represent two distinct objects of interest. To make this distinction concrete, suppose that there exists a finite-dimensional state vector $\mathbf{S}_{t-1}$ that is $\mathcal{G}_{t-1}$ -measurable and $\mu_t(\beta) = \mu_0(\beta; \mathbf{S}_{t-1})$. Further, assume that $\{\mathbf{S}_t\}$ is stationary and ergodic and $\mu_0(\cdot; \cdot)$ satisfies the regularity conditions required for a uniform law of large numbers.¹⁰ Assumption P then holds with $\mu(\beta) = \mathbb{E}[\mu_0(\beta; \mathbf{S}_0)]$. Thus, the PACER averages conditional expected returns over the stationary distribution of the state variables, whereas the SACER averages conditional expected returns over the particular realizations of those state variables observed during the sample period. Consequently, the SACER and PACER may differ meaningfully in any particular sample.

We recommend that empirical researchers focus on the SACER rather than the PACER for a few reasons. First, even if Assumption P fails to hold, the SACER is well-defined under our remaining assumptions. The PACER may not exist if, for example, some state variables driving $\mu_t(\cdot)$ are not stationary. Second, and more importantly, the SACER is a more intuitive and interpretable object of interest as we have a wealth of data summarizing what occurred over our sample. For example, we observe the behavior of macroeconomic aggregates, changes in the legal and regulatory landscape, or the presence of unusual events (e.g., financial crises or natural disasters). Conversely, interpretation of the PACER will generally hinge on the stationary distribution of the economy-wide state variables and the mapping $\mu_0(\cdot; \cdot)$ from those states to conditional expected returns. Characterizing the properties of this distribution would be challenging (e.g., assessing the probability of recessions or financial crises) and the appeal of the approach herein is that we can avoid making specific assumptions about the functional form of $\mu_t(\cdot)$. Third, inference on the PACER requires additional assumptions beyond those needed for inference on the SACER. In addition to the estimation error $\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)$, inference on the PACER must characterize the time-series variation in $\bar{\mu}_T(\beta; H) - \mu(\beta)$. Despite our preference for the SACER, Section 4.3.1 discusses this additional source of uncertainty and the resulting challenges for inference on the PACER.

4 Inference for beta sorted portfolios

In the previous section we established consistency under minimal conditions, requiring only that the first-step estimators are well behaved and the portfolio-formation rule, using these estimators,

¹⁰Equation (2.2) allows for more generality; however, we form our discussion around a finite set of state variables governing the economy for simplicity of exposition and to cement intuition.

results in portfolios with sufficiently similar β values and sufficiently many assets. To move the theory beyond consistency, however, we will now specialize to the family of first-step estimators and portfolio-formation rules that are in widespread practice.

4.1 Classical beta-sorted portfolio estimator

The classical beta-sorted portfolio estimator is comprised of (i) rolling window OLS regressions to estimate β_{it} as given by equation (2.10) and (ii) forming portfolios based on the marginal quantiles of the cross-sectional distribution of $\hat{\beta}_{it}$ as given by equation (2.11) in the special case of $d = 1$. Taking these mainstays of empirical practice as our starting point, we will obtain an asymptotic linearization of $\hat{\mu}(\cdot)$ that underpins our results on asymptotic normality and feasible inference.

The first ingredient of beta sorts is an estimate of β_{it} . It will be convenient to introduce the following generalization of the conventional rolling window estimator. Let $1 \leq s \leq H$ and define

$$r_{\mathbf{p}}(u) = (1, u, \dots, u^{\mathbf{p}})^{\top}, \quad X_s(t) = r_{\mathbf{p}}(s/H) \otimes (1, f_{t-s}^{\top})^{\top}$$

where $\otimes$ denotes the Kronecker product and $\mathbf{p}$ is a fixed integer. The corresponding estimator of β_{it} is formed from the following local polynomial regression estimator,

$$\hat{\beta}_{it} = S \hat{\theta}_{it}, \quad \hat{\theta}_{it} = \left(\sum_{s=1}^H X_s(t) X_s(t)^{\top} \right)^{-1} \sum_{s=1}^H X_s(t) R_{i(t-s)}, \quad (4.1)$$

where S selects elements $2, \dots, d+1$ of $\hat{\theta}_{it}$ which correspond to the estimator of β_{it} . This is still a rolling-window least squares estimator except with higher-order polynomial terms. In SA-Theory we also allow for kernel weights in equation (4.1) but we suppress this generalization since we focus on equal weights to match empirical practice. When $\mathbf{p} = 0$, we recover the rolling-window estimator in equation (2.10). For the SACER estimation and inference procedures discussed in this section, we use $\mathbf{p} = 0$ and $\mathbf{p} = 1$. The general formulation also accommodates the local quadratic regressions, $\mathbf{p} = 2$, used in the empirical affine specification test.

We require that $H \rightarrow \infty$ and $H/T \rightarrow 0$ to ensure that the number of observations in each window grows but the size of the window itself, relative to the full sample, shrinks. The polynomial order, $\mathbf{p}$, controls the flexibility of the fit within the window. A more flexible fit can reduce the

(first-step) approximation bias. Indeed, we will use the first-step estimator with $\mathbf{p} = 1$ to conduct valid pointwise inference at fixed nonzero evaluation points based on a data-driven choice of H (see Section [SA-III.1](#) of SA-Applied for details).¹¹

Just as the first step of the estimation procedure can be generalized to allow for more flexible fits (i.e., $\mathbf{p} > 0$), so too can the second step. For example, rather than use a constant fit within the portfolio we can use the least-squares fit from a regression of R_{it} on a constant and $\hat{\beta}_{it}$ using *only* the assets placed in that portfolio. To distinguish the order of this second-step fit from the first-step polynomial order, let $\mathbf{q}$ denote the polynomial order within each portfolio and write the corresponding estimators as $\widehat{M}_t(\beta; \mathbf{q})$ and $\widehat{\mu}(\beta; \mathbf{q}) = (T - H)^{-1} \sum_{t=H+1}^T \widehat{M}_t(\beta; \mathbf{q})$. The case $\mathbf{q} = 0$ is exactly the classical portfolio-mean estimator whereas $\mathbf{q} = 1$ corresponds to running linear regressions within the portfolio. This more flexible fit can reduce the magnitude of the second-step approximation bias. In particular, we will need to use $\widehat{\mu}(\beta; 1)$ to conduct inference when $\beta = 0$. Further, even when $\beta \neq 0$, using $\mathbf{q} = 1$ can reduce the magnitude of the approximation bias. It also eliminates a higher-order remainder term exactly.¹² Throughout, we use $(\mathbf{p}, \mathbf{q}) = (0, 0)$ for point estimation and $(\mathbf{p}, \mathbf{q}) = (1, 1)$ for inference on the SACER.

4.2 Asymptotic normality

In this section, we establish asymptotic normality of the estimator under the additional conditions collected in Assumption [SA-DGP](#) of SA-Theory. Most are standard regularity conditions, so we discuss only the more noteworthy restrictions here. First, we impose more structure on the idiosyncratic errors by writing $\varepsilon_{it} = \varsigma_{it}\eta_{it}$, requiring the primitive innovations, η_{it} , to be independent across assets and time while allowing for cross-sectional and time-series dependence in the scale factors, ς_{it} (Assumption [SA-DGP\(ii\)](#)). Second, we impose conditions on α_{it} and β_{it} to ensure a well-behaved time-varying coefficient model and assume that the histories of these coefficients through date t are conditionally independent across assets, while allowing their conditional distributions to differ across assets (Assumption [SA-DGP\(iii\)](#)). This permits systematic differences in factor exposures across groups such as industries. Third, we impose a near-epoch dependence assumption on the risk factors and require the time average of the conditional variance to converge to a deterministic

¹¹Section [SA-II.2](#) of SA-Theory also provides additional results, including primitive smoothness conditions and uniform convergence rates for the first-step estimator with arbitrary $\mathbf{p}$.

¹²Specifically, $\mathbf{q} \geq 1$ implies that $\mathcal{R}_3(\beta) = 0$; see Lemma [SA-I.3.1](#) of SA-Theory.

positive-definite matrix (Assumption [SA-DGP\(iv\)](#)).

The starting point for the distributional analysis is the stochastic linearization. For fixed $\mathbf{q} \in \{0, 1\}$, we have that

$$\hat{\mu}(\beta; \mathbf{q}) - \bar{\mu}_T(\beta; H) = \bar{\mathcal{J}}_f(\beta) + \bar{\mathcal{J}}_\varepsilon(\beta; \mathbf{q}) + \mathcal{R}(\beta; \mathbf{q}), \quad (4.2)$$

where $\bar{\mathcal{J}}_f(\beta)$ is the risk factor component,

$$\bar{\mathcal{J}}_f(\beta) = \frac{1}{T-H} \sum_{t=H+1}^T \beta^\top (f_t - \mathbb{E}[f_t | \mathcal{G}_{t-1}]), \quad (4.3)$$

$\bar{\mathcal{J}}_\varepsilon(\beta; \mathbf{q})$ is the idiosyncratic component, and $\mathcal{R}(\beta; \mathbf{q})$ collects all remaining approximation errors. The idiosyncratic component captures two distinct sources of estimation error. The first source is the direct contribution of the idiosyncratic shocks, ε_{it} , to the second-step portfolio fit. This term also arises in standard nonparametric regression problems. The second source is the indirect contribution of past idiosyncratic shocks through the estimated betas used in the second step. Because the rolling estimation windows overlap, the same shock can affect beta estimates in more than one period. Formal expressions are provided in Section [SA-II.4](#) of SA-Theory.

We can use equation (4.2) to introduce some key properties of $\hat{\mu}(\beta; \mathbf{q})$. First, under the rate conditions in Section [SA-II.6](#) of SA-Theory, it can be shown that $\mathcal{R}(\beta; \mathbf{q})$ is negligible relative to the other terms in the linearization. Second, the remaining terms $\bar{\mathcal{J}}_f(\beta)$ and $\bar{\mathcal{J}}_\varepsilon(\beta; \mathbf{q})$ have different rates of convergence. Specifically, $\bar{\mathcal{J}}_f(0) = 0$ and

$$\bar{\mathcal{J}}_f(\beta) = O_{\mathbb{P}}\left(\frac{1}{\sqrt{T}}\right), \quad \beta \neq 0, \quad (4.4)$$

whereas

$$\bar{\mathcal{J}}_\varepsilon(0; \mathbf{q}) = O_{\mathbb{P}}\left(\sqrt{\frac{J^d}{nT}}\right), \quad (4.5)$$

and, importantly, $\bar{\mathcal{J}}_\varepsilon(\beta; \mathbf{q}) = o_{\mathbb{P}}(T^{-1/2})$ when $\beta \neq 0$. Thus, $\bar{\mathcal{J}}_f(\beta)$ is the dominant first-order term when $\beta \neq 0$. This implies that, at any fixed nonzero evaluation point, the estimator achieves only a $\sqrt{T}$ rate of convergence even though it uses approximately nT observations. However, the pointwise convergence rate changes discontinuously at $\beta = 0$. At the origin, $\bar{\mathcal{J}}_f(0) = 0$, so, to first

order, only $\bar{\mathcal{S}}_\varepsilon(0; \mathbf{q})$ contributes, yielding a faster convergence rate.

The $\sqrt{T}$ convergence rate is a consequence of the common factor structure imposed in equation (2.1). At any fixed $\beta \neq 0$, each period's return contains the innovation to the risk factor, $\beta^\top (f_t - \mathbb{E}[f_t \mid \mathcal{G}_{t-1}])$. Increasing the cross-sectional sample size averages away the effect of idiosyncratic shocks but has no effect on this common component, whose influence can only be reduced by averaging over time. It is this feature that explains why $\bar{\mathcal{S}}_f(\beta)$ dominates the other terms in the linearization. This slower rate of convergence is the cost of remaining agnostic about the functional form of $\mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$. If the conditional mean of the risk factors were known, the factor innovation could be removed from each portfolio return and a faster convergence rate could then be achieved. However, this would require a functional-form assumption on the factor dynamics and would lead to an entirely different estimator. As our focus is on the classical beta-sorted portfolio estimator, we do not discuss such an approach further.

The following theorem formalizes the distinction between the pointwise asymptotic distributions at zero and at nonzero evaluation points. Let $\mathcal{B}^\circ \subset \text{int}(\mathcal{B})$ denote an interior evaluation region containing zero.

Theorem 4.1 (Pointwise asymptotic normality). *Fix $\beta \in \mathcal{B}^\circ$ and suppose the conditions of Theorem SA-II.6.2 of SA-Theory hold. If $\beta \neq 0$, then, for each $\mathbf{q} \in \{0, 1\}$,*

$$\frac{\hat{\mu}(\beta; \mathbf{q}) - \bar{\mu}_T(\beta; H)}{\sqrt{\Sigma_f(\beta)}} \rightsquigarrow \mathcal{N}(0, 1), \quad \Sigma_f(\beta) \asymp_{\mathbb{P}} T^{-1},$$

where $\rightsquigarrow$ denotes convergence in distribution and

$$\Sigma_f(\beta) = \frac{1}{(T - H)^2} \sum_{t=H+1}^T \beta^\top \mathbb{V}\text{ar}(f_t \mid \mathcal{G}_{t-1}) \beta.$$

If $\beta = 0$, then

$$\frac{\hat{\mu}(0; 1) - \bar{\mu}_T(0; H)}{\sqrt{\Sigma_\varepsilon(0; 1)}} \rightsquigarrow \mathcal{N}(0, 1), \quad \Sigma_\varepsilon(0; 1) \asymp_{\mathbb{P}} \frac{J^d}{nT},$$

where $\Sigma_\varepsilon(0; \mathbf{q})$ is the conditional variance of $\bar{\mathcal{S}}_\varepsilon(0; \mathbf{q})$ and is defined in Section SA-II.6 of SA-Theory.

Theorem 4.1 establishes pointwise asymptotic normality of the beta-sorted portfolio estimator centered at the SACER and formalizes the distinction between the convergence rates described

above. Although the leading asymptotic variance at any fixed nonzero evaluation point, $\Sigma_f(\beta)$, does not depend on the order $\mathbf{q}$ of the second-step fit, the choice of $\mathbf{q}$ affects the second-step approximation bias, and $\mathbf{q} = 1$ is required to conduct inference at the origin. For all values of β , asymptotic normality also requires the approximation errors from estimating the β_{it} in the first step and fitting $\mu_t(\cdot)$ in the second step to be negligible relative to the corresponding standard error.

Remark 4.2 (Technical details on Theorem 4.1). Section SA-I.3 of SA-Theory develops a stochastic linearization based on generic first-step estimators, and Section SA-II.4 specializes it to the classical estimator outlined in Section 4.1. Further, SA-Theory gives conditions under which $\mathcal{R}(\beta; \mathbf{q})$ is negligible relative to $T^{-1/2}$ when $\beta \neq 0$ and $\sqrt{J^d/(nT)}$ when $\beta = 0$. To establish our pointwise inference result at $\beta = 0$, we modify the marginal-quantile partition in equation (2.11) so that zero is a fixed breakpoint in every marginal partition. This ensures that zero occupies the same boundary position in its selected portfolio at every date, providing the sharper control of the estimated partition needed at the origin. At a fixed $\beta \neq 0$, the leading common risk-factor component, $\bar{\mathcal{S}}_f(\beta)$, does not depend on this adjustment. The partition anchored at zero can nevertheless be used at all evaluation points to provide a common implementation. Importantly, the marginal quantile partition of equation (2.11) remains valid for consistent estimation of the SACER.

Remark 4.3 (Cross-sectional dependence). Equation (4.2) clarifies the scope for weakening the cross-sectional independence assumptions. For a fixed $\beta \neq 0$, the leading term is $\bar{\mathcal{S}}_f(\beta)$, whose behavior is solely governed by the evaluation point and the time-series properties of the risk factors. Consequently, sufficiently weak cross-sectional dependence that preserves the orders of $\bar{\mathcal{S}}_\varepsilon(\beta; \mathbf{q})$ and $\mathcal{R}(\beta; \mathbf{q})$ would leave the first-order distribution unchanged. This suggests that the result could accommodate, for example, additional weak cross-sectional dependence in the idiosyncratic errors of the type considered by Chamberlain and Rothschild (1983).

Remark 4.4 (Time series properties of conditional expected returns). One way to satisfy the first-step bias requirement in Assumption SA-FSB of SA-Theory is to require α_{it} and β_{it} to vary sufficiently smoothly within each rolling estimation window. However, equation (2.4) shows that the relevant restriction cannot be interpreted independently of the functional form for conditional expected returns because $\alpha_t(\beta) = \mu_t(\beta) - \beta^\top \mathbb{E}[f_t | \mathcal{G}_{t-1}]$. To illustrate, suppose that $\mathbf{S}_t$ is a vector of $\mathcal{G}_t$ -measurable state variables which follows a stationary linear time-series process (with nonde-

generate innovations) and $\mathbb{E}[f_t \mid \mathcal{G}_{t-1}] = \zeta + \Lambda \mathbf{S}_{t-1}$, with $\Lambda \neq 0$ (as in, e.g., [Adrian et al. 2015](#) or [Gagliardini et al. 2016](#)). Then, in order for $\alpha_{it} = \alpha_t(\beta_{it})$ to vary smoothly, $\mu_t(\cdot)$ *must* vary over time in a way that offsets the nonsmooth variation induced by $\beta^\top \Lambda \mathbf{S}_{t-1}$. In that case, the SACER and PACER are distinct objects and need not coincide in a given sample, which directly affects the interpretation of estimation and inference in empirical applications.

4.3 Feasible inference on the SACER

We have established conditions which ensure asymptotic normality of the beta-sorted portfolio estimator when centered at the SACER. In this section, we investigate the properties of two feasible inference procedures. The first is the so-called Fama-MacBeth (FM) variance estimator which is ubiquitous in existing applications of beta-sorted portfolios. The FM variance estimator can be motivated by the classical sample variance estimator and constructed as

$$\widehat{\Sigma}_{\text{FM}}(\beta; \mathbf{q}) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \left(\widehat{M}_t(\beta; \mathbf{q}) - \widehat{\mu}(\beta; \mathbf{q}) \right)^2. \quad (4.6)$$

Here, $\widehat{M}_t(\beta; \mathbf{q})$ for $t = H+1, \dots, T$ serves as the sample “observations” and $\widehat{\mu}(\beta; \mathbf{q})$ serves as the sample “mean.” Consequently, apart from a degrees-of-freedom correction, the FM variance estimator also has an interpretation as a leave- t out jackknife variance estimator. $\widehat{\Sigma}_{\text{FM}}(\beta; 0)$ is ubiquitous in the empirical finance literature. We will discuss the theoretical properties of this conventional FM variance estimator, and its generalization given by equation (4.6), later in this section.

The second feasible inference procedure is based on the theoretical results we have presented. From equation (4.2) and Theorem 4.1, sampling uncertainty has a common risk-factor component, $\Sigma_f(\beta)$, and an idiosyncratic component, $\Sigma_\varepsilon(\beta; 1)$, which we will estimate separately. At any fixed $\beta \neq 0$, $\Sigma_f(\beta)$ is the leading variance component, which we estimate using the following plug-in estimator:

$$\widehat{\Sigma}_f(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \left[\beta^\top (f_t - \widehat{f}_{t|t-1}) \right]^2. \quad (4.7)$$

Here, $\widehat{f}_{t|t-1}$ is an estimate of $\mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$ using only information contained in $\mathcal{G}_{t-1}$. For example, a parametric approach would lead to $\widehat{f}_{t|t-1} = h_{t-1}(\widehat{\vartheta})$ for some $h_{t-1}(\vartheta)$ with $h_{t-1}(\widehat{\vartheta})$ a corresponding

feasible estimate. As a simple example, in the case of an AR(1) we would obtain $h_{t-1}(\vartheta) = \vartheta_0 + \vartheta_1 f_{t-1}$, and we would estimate the parameters, $(\vartheta_0, \vartheta_1)$, recursively. Importantly, the estimator, $\hat{f}_{t|t-1}$, need not be based on the correct specification. If the conditional mean of the factors is misspecified, then Theorem 4.5 below shows that

$$\hat{\Sigma}_f(\beta) - \Sigma_f(\beta) = \mathcal{B}_F(\beta) + o_{\mathbb{P}}(T^{-1}), \quad (4.8)$$

where

$$\mathcal{B}_F(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \left[\beta^\top (\hat{f}_{t|t-1} - \mathbb{E}[f_t | \mathcal{G}_{t-1}]) \right]^2. \quad (4.9)$$

Clearly, $\mathcal{B}_F(\beta)$ is non-negative; thus, we can still conduct valid, albeit possibly conservative, inference based on $\hat{\Sigma}_f(\beta)$.

We estimate the variance contribution from the idiosyncratic component using a bin-restricted delete- i jackknife. Let $\widehat{M}_t^{(-i)}(\beta; \mathbf{q})$ be the period- t portfolio estimator computed after deleting unit i from the second-step cross-section while holding the portfolio formation rule represented by the partition $\hat{\Delta}_t$ fixed, and define

$$\hat{d}_{it}(\beta; \mathbf{q}) = \widehat{M}_t^{(-i)}(\beta; \mathbf{q}) - \widehat{M}_t(\beta; \mathbf{q}). \quad (4.10)$$

Set $\hat{d}_{it}(\beta; \mathbf{q}) = 0$ when unit i is not observed at date t . The jackknife variance estimator is

$$\hat{\Sigma}_\varepsilon(\beta; \mathbf{q}) = \frac{1}{(T-H)^2} \sum_i \sum_{t=H+1}^T \sum_{s=H+1}^T \hat{d}_{it}(\beta; \mathbf{q}) \hat{d}_{is}(\beta; \mathbf{q}) \mathbb{1}\{|t-s| \leq H\}. \quad (4.11)$$

The truncation at lag H captures the serial dependence induced by the overlapping first-step estimation windows. Pairs farther than H periods apart do not share primitive idiosyncratic shocks after the first-step correction is isolated. Importantly, calculating $\hat{\Sigma}_\varepsilon(\beta; \mathbf{q})$ does not require any re-estimation steps in practice. Using properties of leave-one-out OLS estimation, we can express $\hat{d}_{it}(\beta; \mathbf{q})$ as a function of the fitted residuals from the second-step regression and the leverage of unit i in that regression. Section SA-II.7 of SA-Theory gives equivalent computationally simple formulations of this jackknife variance estimator.

Combining the two components gives our proposed variance estimator

$$\widehat{\Sigma}(\beta) = \widehat{\Sigma}_f(\beta) + \widehat{\Sigma}_\varepsilon(\beta; 1). \quad (4.12)$$

Theorem 4.5 (Plug-in variance estimator). *Fix $\beta \in \mathcal{B}^\circ$ and suppose the conditions of Theorem SA-II.7.3 of SA-Theory hold. If $\beta \neq 0$, then*

$$\widehat{\Sigma}(\beta) - \Sigma_f(\beta) - \Sigma_\varepsilon(\beta; 1) - \mathcal{B}_F(\beta) = o_{\mathbb{P}}(T^{-1}).$$

If $\beta = 0$, then

$$\widehat{\Sigma}(0) - \Sigma_\varepsilon(0; 1) = o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

Theorem 4.5 shows that the newly-proposed variance estimator satisfies the required rate restrictions to ensure feasible inference. When the conditional mean of the risk factors is correctly specified and estimated sufficiently accurately, the plug-in variance estimator is consistent for the first-order variance. When the conditional mean is misspecified, the plug-in variance estimator is conservative and overestimates the first-order variance by $\mathcal{B}_F(\beta)$. The jackknife variance estimator is consistent for the idiosyncratic component of the first-order variance at $\beta = 0$. Thus, Theorem 4.5 establishes that $\widehat{\Sigma}(\beta)$ for $\beta \neq 0$ and $\widehat{\Sigma}(0)$ at the origin can be used to conduct valid, possibly conservative, pointwise inference.

We now discuss the properties of the Fama-MacBeth variance estimator.

Theorem 4.6 (Fama-MacBeth variance estimator). *Fix $\beta \in \mathcal{B}^\circ$ and $\mathbf{q} \in \{0, 1\}$, and suppose the conditions of Theorem SA-II.7.4 of SA-Theory hold. Define*

$$\Sigma_\mu(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T (\mu_t(\beta) - \bar{\mu}_T(\beta; H))^2.$$

If $\beta \neq 0$, then

$$\widehat{\Sigma}_{\text{FM}}(\beta; \mathbf{q}) = \Sigma_f(\beta) + \Sigma_\mu(\beta) + o_{\mathbb{P}}(T^{-1}).$$

Theorem 4.6 shows that the FM variance estimator is generally conservative for inference on the SACER at a fixed nonzero β as it includes the time-series variation in conditional expected returns,

$\Sigma_\mu(\beta)$, which is not part of the first-order variance. This additional term is nonnegative and can be strictly positive when $\mu_t(\beta)$ varies over time. The intuition for this result can be gleaned from the following decomposition of the summands of the estimator:

$$\widehat{M}_t(\beta; \mathbf{q}) - \widehat{\mu}(\beta; \mathbf{q}) = [\widehat{M}_t(\beta; \mathbf{q}) - \mu_t(\beta)] + [\mu_t(\beta) - \bar{\mu}_T(\beta; H)] - [\widehat{\mu}(\beta; \mathbf{q}) - \bar{\mu}_T(\beta; H)]. \quad (4.13)$$

After squaring and averaging this decomposition, the first component generates the leading sampling variance, $\Sigma_f(\beta)$, while the second generates $\Sigma_\mu(\beta)$. The final component and the relevant cross-products are asymptotically negligible.

The properties of the FM variance estimator at $\beta = 0$ are very different. As we saw in Section 4.2, the uncertainty of the estimator depends importantly on the fact that the second-step procedure relies on first-step estimation. The FM estimator uses the period-by-period dispersion of the portfolio estimates but does not account for the dependence induced by the overlapping windows in the first step. As such, it need not estimate the appropriate variance at the origin. This is the case even when the no-arbitrage condition holds and the conditional expected return function is linear in β . Depending on the DGP, $\widehat{\Sigma}_{\text{FM}}(0; \mathbf{q})$ may either overestimate or underestimate the true variance.

Corollary 4.6.1 (Pointwise feasible inference). *Suppose the conditions of Theorems 4.1, 4.5, and 4.6 hold. Let $\mathbf{z}_{1-\alpha/2}$ be the $1 - \alpha/2$ quantile of a standard Gaussian random variable. For any nonnegative variance estimator $\widehat{V}$, define*

$$\mathcal{I}_{1-\alpha}(\beta; \mathbf{q}, \widehat{V}) = \left[\widehat{\mu}(\beta; \mathbf{q}) - \mathbf{z}_{1-\alpha/2} \widehat{V}^{1/2}, \widehat{\mu}(\beta; \mathbf{q}) + \mathbf{z}_{1-\alpha/2} \widehat{V}^{1/2} \right]$$

and $\widehat{\Sigma}(\beta)_+ = \max \{ \widehat{\Sigma}(\beta), 0 \}$. Then:

(i) For each fixed $\beta \neq 0$ and $\mathbf{q} \in \{0, 1\}$,

$$\liminf_{n, T \rightarrow \infty} \mathbb{P} \left(\bar{\mu}_T(\beta; H) \in \mathcal{I}_{1-\alpha}(\beta; \mathbf{q}, \widehat{\Sigma}(\beta)_+) \right) \geq 1 - \alpha.$$

The coverage probability converges to $1 - \alpha$ if $T\mathcal{B}_{\text{F}}(\beta) = o_{\mathbb{P}}(1)$.

(ii) For each fixed $\beta \neq 0$ and $\mathbf{q} \in \{0, 1\}$,

$$\liminf_{n, T \rightarrow \infty} \mathbb{P} \left(\bar{\mu}_T(\beta; H) \in \mathcal{I}_{1-\alpha}(\beta; \mathbf{q}, \hat{\Sigma}_{\text{FM}}(\beta; \mathbf{q})) \right) \geq 1 - \alpha,$$

The coverage probability converges to $1 - \alpha$ if $T\Sigma_\mu(\beta) = o_{\mathbb{P}}(1)$.

(iii) For $\beta = 0$,

$$\mathbb{P} \left(\bar{\mu}_T(0; H) \in \mathcal{I}_{1-\alpha}(0; 1, \hat{\Sigma}(0)_+) \right) \rightarrow 1 - \alpha.$$

Corollary 4.6.1 shows that for $\beta \neq 0$ confidence intervals based on either the newly-proposed variance estimator or the FM variance estimator attain at least their nominal coverage probability. The former variance estimator is asymptotically exact when $T\mathcal{B}_{\text{F}}(\beta)$ is asymptotically negligible, whereas the FM interval is asymptotically exact when the time-series variation in conditional expected returns is asymptotically negligible. When $\beta = 0$, the newly-proposed variance estimator is asymptotically exact. The use of $\hat{\Sigma}(\beta)_+$ in Corollary 4.6.1 arises because the jackknife component of the variance estimator, $\hat{\Sigma}_\varepsilon(\beta; \mathbf{q})$, can be negative in finite samples.¹³ This ensures that the standard error is well defined. In the limit, this adjustment is unnecessary with probability approaching one.

Remark 4.7 (Uniformity over β). Corollary 4.6.1 establishes coverage at each fixed evaluation point, but it does not guarantee that coverage remains valid uniformly as β approaches zero. This distinction matters because of the differential behavior of the limiting distribution at the origin that we have discussed. Section SA-II.8 of SA-Theory develops a uniformly valid procedure that uses the newly-proposed variance estimator away from zero and a conservative variance adjustment near zero. Let $\mathcal{I}_{1-\alpha}^{\text{UI}}(\beta)$ denote the resulting pointwise confidence interval. These confidence intervals attain at least their nominal coverage probability uniformly over $\beta \in \mathcal{B}^\circ$, i.e.,

$$\liminf_{n, T \rightarrow \infty} \inf_{\beta \in \mathcal{B}^\circ} \mathbb{P} \left(\bar{\mu}_T(\beta; H) \in \mathcal{I}_{1-\alpha}^{\text{UI}}(\beta) \right) \geq 1 - \alpha.$$

Remark 4.8. Theorems 4.5 and 4.6 imply that we can construct an asymptotic lower bound on

¹³Alternatively, we could add weights to the jackknife variance estimator to ensure non-negativity as is done in the HAC literature (e.g., Newey and West, 1987). However, in empirically-relevant settings we do not observe negative values of $\hat{\Sigma}_\varepsilon(\beta; \mathbf{q})$.

the time-series variation in conditional expected returns, as measured by $\Sigma_\mu(\beta)$, by taking the difference between $\widehat{\Sigma}_{\text{FM}}(\beta; \mathbf{q})$ and $\widehat{\Sigma}(\beta)$. Under the assumptions of these theorems, for each fixed $\beta \neq 0$ and $\mathbf{q} \in \{0, 1\}$,

$$\widehat{\Sigma}_{\text{FM}}(\beta; \mathbf{q}) - \widehat{\Sigma}(\beta) = \Sigma_\mu(\beta) - \mathcal{B}_{\text{F}}(\beta) + o_{\mathbb{P}}(T^{-1}). \quad (4.14)$$

Because $\mathcal{B}_{\text{F}}(\beta) \geq 0$, equation (4.14) shows that this difference provides an asymptotic lower bound on $\Sigma_\mu(\beta)$. When $T\mathcal{B}_{\text{F}}(\beta) = o_{\mathbb{P}}(1)$, the bound is asymptotically sharp. Thus, the contrast between the two variance estimators provides information about the time-series variation in conditional expected returns. In both our empirical applications, we find that this difference is relatively large, making the bound informative.

Remark 4.9. Theorem 4.5 and Corollary 4.6.1 imply that valid, although potentially conservative, inference on the SACER does not require correct specification of $\mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$. A misspecified predictor increases the estimated variance through $\mathcal{B}_{\text{F}}(\beta)$ but does not invalidate coverage. This contrasts with, for example, Adrian et al. (2015) and Gagliardini et al. (2016), which impose a first-order Markov structure on the factor dynamics. In practice, specifying the correct functional form and the appropriate conditioning variables for the factor dynamics can be challenging. This robustness is an advantage of the beta-sorted portfolio estimator.

The choices of H and J govern the amount of smoothing in the two estimation steps. Longer first-step windows reduce the variability of the estimated betas but can increase approximation bias when risk-factor exposures vary over time, while forming more portfolios reduces within-portfolio approximation bias but leaves fewer assets in each portfolio. Part III of SA-Applied develops data-driven choices, based on mean squared error criteria, at fixed nonzero evaluation points, together with a near-origin adjustment for uniform-in- β inference. It also explains why factor-cancelling inference requires dedicated undersmoothing. Part IV reports Monte Carlo evidence on the finite-sample performance of the SACER estimators, the proposed and FM variance estimators, and the data-driven tuning rules. The simulation results are consistent with our theoretical predictions. We find that confidence intervals based on our proposed variance estimator perform well when the evaluation point is away from the origin. Near zero, the uniform-in- β adjustment substantially improves coverage relative to the pointwise procedure.

4.3.1 Estimation and Inference on the PACER

Although we recommend focusing on the SACER, it is useful to compare the requirements for estimation and inference on the PACER. For $\mathbf{q} \in \{0, 1\}$, decompose

$$\hat{\mu}(\beta; \mathbf{q}) - \mu(\beta) = [\hat{\mu}(\beta; \mathbf{q}) - \bar{\mu}_T(\beta; H)] + [\bar{\mu}_T(\beta; H) - \mu(\beta)]. \quad (4.15)$$

The first term is $o_{\mathbb{P}}(1)$ under the conditions in Section 3 and Theorem SA-I.2.1 of SA-Theory. At any fixed β , consistency for the PACER therefore requires only a law of large numbers for $\bar{\mu}_T(\beta; H)$. Assumption P imposes this convergence uniformly over $\beta \in \mathcal{B}$.

Inference on the PACER, however, requires substantially more structure. When the estimator is centered at the SACER, its leading component has a martingale-difference structure that underlies Theorem 4.1. Centering at the PACER introduces the additional term $\bar{\mu}_T(\beta; H) - \mu(\beta)$, which reflects the time-series variation in conditional expected returns and is not generally a martingale difference. A PACER central limit theorem must, therefore, characterize the asymptotic distribution of this time-series average, including its long-run variance, as well as its covariance with the SACER-centered estimation error (the first term in equation (4.15)). Assumption P alone is insufficient for such results.

Consequently, the current variance results do not by themselves provide feasible inference on the PACER. At a fixed nonzero β , Theorem 4.6 shows that the FM variance estimator captures the period-by-period variation in conditional expected returns through $\Sigma_{\mu}(\beta)$. It does not, in general, estimate the long-run variance of this process when conditional expected returns are serially dependent, nor does it account for their possible covariance with the SACER-centered estimation error. Establishing a PACER central limit theorem and constructing a feasible variance estimator would therefore require substantially more structure on the time-series dynamics of the DGP.

5 Extensions and further discussion

5.1 Multiple factors and portfolio construction

The beta-sorted portfolio estimator that we have studied throughout the paper is the one based on so-called independent sorts, i.e., forming portfolios based on tensor products of the marginal

partitions of each factor exposure (using marginal quantiles as breakpoints) as described in earlier sections. This is the multivariate analog of the estimator when $d = 1$. A drawback of this approach is that, as we showed, the number of portfolios grows exponentially in the number of factors (at the rate J^d), which can lead to sparsely populated portfolios and even empty portfolios. In practice, researchers often use a small number of factors to avoid this problem; in particular, applications with $d > 2$ are rare (one exception is [Chang et al. \(2013\)](#), who used $d = 4$).

An alternative approach that avoids multidimensional sorting is to form portfolios based on a single target risk-factor exposure, while controlling for the other factors in the first-step regression.¹⁴ However, such an approach targets a different object than the SACER and the distinction is analogous to the difference between the “short” and “long” regression when studying omitted variables in OLS regressions. Suppose that the first risk factor is the target and is used for portfolio construction whereas the other $d - 1$ factors serve only as control variables in the first-step regression. Partition the evaluation point β as $\beta = (\beta_1, \beta_{-1}^\top)^\top$. For simplicity of presentation, suppose the β_{it} are conditionally *i.i.d.* across assets rather than just conditionally independent. Let $\mathbb{E}_t^\beta[\cdot \mid \beta_{it,1} = b_1]$ denote the expectation with respect to their common period- t conditional distribution, further conditioning on the first component being equal to b_1 .¹⁵ The resultant estimator targets the following object

$$\bar{\mu}_{T|1}(b_1; H) = \frac{1}{T - H} \sum_{t=H+1}^T \mu_{t|1}(b_1), \quad \mu_{t|1}(b_1) = \mathbb{E}_t^\beta[\mu_t(\beta_{it}) \mid \beta_{it,1} = b_1]. \quad (5.1)$$

Like the SACER, $\bar{\mu}_{T|1}(b_1; H)$ is generally random because both conditional expected returns and the cross-sectional distribution of the remaining betas, $\beta_{it,-1}$, may vary over time. Sorting on a single target variable, therefore, averages over the remaining exposures rather than holding them fixed and does not generally recover the SACER at any specified value of $(b_1, b_{-1}^\top)^\top$.

We can use the theoretical structure introduced in earlier sections to describe the asymptotic distribution of this marginal sorting estimator which we label as $\hat{\mu}_{|1}(b_1; \mathbf{q})$. Define the average beta of assets with target risk factor $\beta_1 = b_1$ at time t as $\vartheta_{t|1}(b_1) = \mathbb{E}_t^\beta[\beta_{it} \mid \beta_{it,1} = b_1]$. Under corresponding regularity and rate conditions, the leading expansion of the marginal beta-sorted

¹⁴For recent examples, see [Bali et al. \(2017\)](#), [Chen et al. \(2021\)](#), [Fan et al. \(2022\)](#), and [Chen et al. \(2023\)](#).

¹⁵Formally, $\mathbb{E}_t^\beta[\cdot \mid \beta_{it,1} = b_1] = \mathbb{E}[\cdot \mid \mathcal{D}_{t-1}, \beta_{it,1} = b_1]$ where $\mathcal{D}_{t-1}$ is the design sigma field which we suppress for simplicity. The design sigma field and the related filtrations are defined in Section [SA-II.1.2](#) of SA-Theory.

portfolio estimator is

$$\hat{\mu}_{|1}(b_1; \mathbf{q}) - \bar{\mu}_{T|1}(b_1; H) = \frac{1}{T-H} \sum_{t=H+1}^T \vartheta_{t|1}(b_1)^\top (f_t - \mathbb{E}[f_t | \mathcal{G}_{t-1}]) + \bar{\mathcal{J}}_{|1}(b_1) + \mathcal{R}_{|1}(b_1), \quad (5.2)$$

where $\bar{\mathcal{J}}_{|1}(b_1) = O_{\mathbb{P}}(\sqrt{J/(nT)})$ represents the contribution from the idiosyncratic component and $\mathcal{R}_{|1}(b_1)$ collects all other remainder terms. As before, when the risk factor component is nondegenerate (and the remainder is negligible) the estimator possesses a $\sqrt{T}$ rate. Importantly, setting $b_1 = 0$ does not generally eliminate the factor component because the average exposures to the remaining factors may be nonzero. The stronger condition $\vartheta_{t|1}(0) = 0$ for all t eliminates the common-factor component at $b_1 = 0$. Our theoretical framework could be extended to conduct inference on $\bar{\mu}_{T|1}(b_1; H)$ using a variance estimator analogous to $\widehat{\Sigma}(\beta)$. However, this marginal target depends on the joint distribution of risk-factor exposures and can change even when the conditional expected return function remains unchanged. Thus, care must be taken when translating empirical results based on marginal sorting to economic conclusions.

A more principled alternative for mitigating the sparse portfolio problem is to impose additive separability of conditional expected returns across the components of β . Specifically, suppose that

$$\mu_t(\beta) = \mu_{0t} + \sum_{k=1}^d \mu_{kt}(\beta_k), \quad \mu_{kt}(0) = 0, \quad k = 1, \dots, d. \quad (5.3)$$

Additive separability rules out interactions among factor exposures because the effect of changing β_k cannot depend on the remaining components of β . However, it still retains substantial flexibility as each $\mu_{kt}(\cdot)$ may be nonlinear and time varying. Notice that the no-arbitrage restriction is nested as the special case where $\mu_{kt}(\beta_k) = \beta_k \lambda_{kt}$.

Under additive separability, a lower-dimensional estimator is available. To provide intuition, consider the case where $\mathbf{q} = 0$. Without additive separability, $\widehat{M}_t(\cdot; 0)$ can be obtained, at each t , via a cross-sectional regression of R_{it} on approximately J^d indicator functions. In contrast, additive separability of $\mu_t(\cdot)$ directly implies additive separability of the realized systematic return function $M_t(\cdot)$. We can therefore rely on a cross-sectional regression of R_{it} on approximately dJ indicator functions instead.¹⁶

¹⁶A fully additively separable specification is not necessary. Partially additive specifications can allow interactions within low-dimensional groups of risk factor exposures while retaining much of the reduction in dimensionality. In

Importantly, additive separability alone does not imply that the marginal sorting estimator discussed above recovers the component $\mu_{1t}(b_1)$. Under the additive model in equation (5.3),

$$\mu_{t|1}(b_1) = \mu_{0t} + \mu_{1t}(b_1) + \sum_{k=2}^d \mathbb{E}_t^\beta[\mu_{kt}(\beta_{it,k}) \mid \beta_{it,1} = b_1]. \quad (5.4)$$

When the risk-factor exposures are dependent, each summand in the final term can vary with b_1 , which can alter both the level and the shape of $\mu_{t|1}(b_1)$. In the special case in which each summand is (almost surely) constant in b_1 , marginal sorting recovers the shape and contrasts of $\mu_{1t}(b_1)$. However, neither zero correlation nor zero rank correlation across the dimensions of risk-factor exposures is sufficient for this conclusion. Joint additive estimation avoids this restriction by controlling flexibly for the other risk-factor exposure components.

A further useful feature of the additive estimator is that it retains a well-defined interpretation when additive separability is misspecified. At each t , the conditional expected return target becomes the best mean-square approximation to $\mu_t(\cdot)$ within the space of additively separable functions, where mean-square error is computed using $\mathbb{E}_t^\beta[\cdot]$. This interpretation follows because the factor component of $M_t(\cdot)$ is already linear in β (and therefore already additively separable). The time-averaged estimator can thus be understood as targeting the sample average of these additive approximations rather than the unrestricted SACER.

Remark 5.1 (Conditional sorts). Another common approach to portfolio construction is a conditional, or sequential, sort.¹⁷ For example, to study exposure β_1 while controlling for exposure β_2 , researchers first sort assets into portfolios using the marginal quantiles of $\hat{\beta}_{it,2}$ and then, within each of those portfolios, sort assets using *conditional* quantiles of $\hat{\beta}_{it,1}$. The procedure can be repeated recursively with additional risk-factor exposures. Because the breakpoints at each stage depend on portfolio membership at the preceding stages, the resulting partition is not a tensor product of marginal-quantile partitions and generally depends on the ordering of the procedure. However, the complete collection of final portfolios generated by the sequential sorts partitions the

that case, the dimensionality cost is governed by the largest group of exposures allowed to interact rather than by the full dimension d . For example, we could have $\mu_t(\beta) = \mu_{0t} + \mu_{12t}(\beta_1, \beta_2) + \sum_{k=3}^d \mu_{kt}(\beta_k)$ so the dimension would be of the order $J^2 + (d-2)J$.

¹⁷For examples of conditional sorts involving estimated risk-factor exposures, see [Herskovic et al. \(2016\)](#), [Bollerslev et al. \(2016\)](#), and [Bali et al. \(2017\)](#).

observed cross-section.¹⁸ Thus, provided that the resulting cells shrink appropriately and their boundaries within the support are recorded, an estimator evaluated at a fixed β continues to target the SACER. Importantly, this statement requires keeping track of the part of the support each portfolio represents. For example, the fifth β_1 portfolio within a low- β_2 group can cover a very different range of β_1 from the fifth β_1 portfolio within a high- β_2 group. If instead portfolio returns or high-minus-low spreads are averaged across parent portfolios based only on their sequential ranks, without explicitly keeping track of the β ranges represented by each portfolio, as is common in empirical applications, the target becomes a weighted average of conditional expected returns or return differences, respectively, and does not generally identify the SACER or a SACER contrast at a fixed beta.

Although conditional sorts can ensure nonempty portfolios when the cross-section is sufficiently large, they do not eliminate the curse of dimensionality. With J portfolios formed for each of d exposures, there are still J^d final portfolios containing approximately n_t/J^d assets each. Thus, conditional sorts alleviate empty portfolios caused by dependence among the risk-factor exposures, but not the sparsity caused by increasing the number of exposures. Our high-level framework can accommodate conditional sorts, but providing primitive conditions under which our results apply is challenging because the recursively estimated breakpoints require, throughout the sequential sorting procedure, uniform control over both the β ranges represented by the portfolios and the number of assets they contain.

5.2 Factor-cancelling portfolios and linear-pricing restrictions

The stochastic expansion in Section 4.2 shows that, for $\beta \neq 0$, the common risk-factor component is the leading source of uncertainty for the beta-sorted portfolio estimator. Inspecting equation (4.3), we observe that this term is linear in β ; consequently, it can be eliminated for judiciously chosen linear contrasts. Let $\mathbf{b}_1, \dots, \mathbf{b}_{K_\Psi}$ be fixed exposure vectors, let $a_1, \dots, a_{K_\Psi}$ be fixed weights, and define $\hat{\Psi}(\mathbf{q}) = \sum_{k=1}^{K_\Psi} a_k \hat{\mu}(\mathbf{b}_k; \mathbf{q})$ and its target $\bar{\Psi}_T = \sum_{k=1}^{K_\Psi} a_k \bar{\mu}_T(\mathbf{b}_k; H)$. If the weights and

¹⁸Although the final portfolios partition the observed cross-section, covering the entire support $\mathcal{B}$ requires extending the outer conditional intervals within every parent portfolio to the corresponding boundaries of $\mathcal{B}$.

evaluation points satisfy

$$\sum_{k=1}^{K_\Psi} a_k \mathbf{b}_k = 0, \quad (5.5)$$

then the leading common risk-factor component in the stochastic expansion cancels exactly.

A specific example of such a linear contrast is a “butterfly” trade where one goes long one unit of each of two assets, one with risk-factor exposure $\mathbf{b}_1$ and one with risk-factor exposure $\mathbf{b}_3$, and short two units of an asset with risk-factor exposure $\mathbf{b}_2$. Then, if $\mathbf{b}_1 - \mathbf{b}_2 = \mathbf{b}_2 - \mathbf{b}_3$ and we take weights as $(1, -2, 1)$, equation (5.5) is satisfied. The estimator for the sample average of the conditional expected returns of such a trade is then

$$\widehat{\Psi}_{\text{BF}}(\mathbf{q}) = \widehat{\mu}(\mathbf{b}_1; \mathbf{q}) - 2\widehat{\mu}(\mathbf{b}_2; \mathbf{q}) + \widehat{\mu}(\mathbf{b}_3; \mathbf{q}). \quad (5.6)$$

This contrast is a discrete second difference and therefore a measure of curvature in the sample-average conditional expected return function.

Consistency of $\widehat{\Psi}(\mathbf{q})$ follows directly under the conditions in Section 3 and Theorem SA-I.2.1 of SA-Theory because the contrast is just a fixed finite linear combination of consistent estimators. In fact, consistency holds regardless of whether equation (5.5) is satisfied. Inference, on the other hand, does not follow directly from the pointwise results in Section 4.2. Because the factor-cancelling restriction eliminates the leading common risk-factor component, the distributional properties of $\widehat{\Psi}(\mathbf{q}) - \bar{\Psi}_T$ in the limit are akin to those of $\widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H)$ when $\beta = 0$.¹⁹ The rate of convergence is now governed by the idiosyncratic component in equation (4.2). Under the conditions in Theorem SA-II.9.1 of SA-Theory, it is shown that the estimator enjoys the faster $\sqrt{nT/J^d}$ rate. In order to derive these properties, however, we require a partition anchored at the prespecified evaluation points, as detailed in Section SA-II.9 of SA-Theory. This construction is the analogue, for factor-cancelling linear contrasts, of the partition anchored at zero (see Remark 4.2).

A natural hypothesis of interest is whether conditional expected returns satisfy the restrictions implied by the absence of arbitrage opportunities. As we have discussed, no-arbitrage conditions

¹⁹Ordinary high-minus-low portfolios do not generally satisfy condition (5.5) and therefore retain the common risk-factor component.

restrict the form of $\mu_t(\beta)$ to be linear (for excess returns) or affine (for raw returns). One approach would be to test whether the time-averaged conditional expected return of a butterfly trade equals zero. A broader approach tests the linear or affine specification jointly across a collection of values for the risk-factor exposures, which is the approach we propose.

Here, we will focus on the affine case. Consider the following null hypothesis,

$$H_0^{\text{LIN}} : \mu_t(b) = a_t + c_t^\top b, \quad b \in \mathcal{B}, \quad t = 1, \dots, T, \quad (5.7)$$

where the intercept and slopes may vary over time. Under the null, realized returns satisfy

$$R_{it} = a_t + \varpi_t^\top \beta_{it} + \varepsilon_{it}, \quad (5.8)$$

where $\varpi_t = c_t + f_t - \mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$. Thus, the realized cross-sectional slope is ϖ_t rather than c_t because it absorbs the contemporaneous risk-factor innovation. If the β_{it} were known, then a_t and ϖ_t could be estimated by a cross-sectional regression at each t .

The procedure evaluates equation (5.7) at a fixed collection of values for the risk-factor exposures: $\mathbf{b}_1, \dots, \mathbf{b}_{K_{\text{LIN}}}$. At each date t , we construct two beta estimates using adjacent, nonoverlapping first-step windows. The earlier estimate, $\hat{\beta}_{it}$, uses observations dated $t - H, \dots, t - 1$, and the later estimate, $\hat{\beta}_{i(t+H)}$, uses observations dated $t, \dots, t + H - 1$ and is paired with the outcome $R_{i(t+H)}$.

Because the later beta estimate contains first-step estimation error, an OLS regression of $R_{i(t+H)}$ on $\hat{\beta}_{i(t+H)}$ would generally exhibit errors-in-variables bias. To address this bias in the linear fit used for the test, we estimate the relation by instrumental variables, using $\hat{\beta}_{it}$ as an instrument for $\hat{\beta}_{i(t+H)}$. Since the two windows do not overlap, the instrument is predetermined with respect to the innovations entering the later beta estimate and asset return. Relevance additionally requires that the earlier beta estimate retain sufficient cross-sectional information about the later beta. This condition is natural in our setting because risk-factor exposures would be expected to evolve gradually over short horizons, preserving the bulk of the cross-sectional information across adjacent estimation windows.

Let $\hat{a}_{t+H}$ and $\hat{\omega}_{t+H}$ denote the resulting IV coefficient estimates and define

$$\hat{e}_{i(t+H)} = R_{i(t+H)} - \hat{a}_{t+H} - \hat{\omega}_{t+H}^\top \hat{\beta}_{i(t+H)}. \quad (5.9)$$

For an evaluation point b , let $\hat{w}_{it}(b)$ denote the weight assigned to asset i by the beta-sorted portfolio estimator using the beta estimates from the earlier window. The test averages the residual returns as follows:

$$\hat{\nu}(b) = \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(b) \hat{e}_{i(t+H)}, \quad (5.10)$$

where $\mathcal{I}_t^{\text{LIN}}$ contains the assets observed in both windows and whose return, $R_{i(t+H)}$, is non-missing. The IV regression, portfolio partition, and normalization of portfolio weights all use this matched sample. Under the null, these residual portfolio returns should be close to zero.

Stacking the moments over the fixed grid gives $\hat{\boldsymbol{\nu}}_{\text{LIN}} = (\hat{\nu}(\mathbf{b}_1), \dots, \hat{\nu}(\mathbf{b}_{K_{\text{LIN}}}))^\top$. The Wald statistic is $\mathbf{W}_{\text{LIN}} = \hat{\boldsymbol{\nu}}_{\text{LIN}}^\top \hat{\Sigma}_{\text{LIN}}^{-1} \hat{\boldsymbol{\nu}}_{\text{LIN}}$, where $\hat{\Sigma}_{\text{LIN}}$ accounts for dependence across the grid points and the serial dependence from overlapping observations. Under the null, $\mathbf{W}_{\text{LIN}}$ converges to a chi-squared distribution with K_{LIN} degrees of freedom, so large values provide evidence against the affine specification at one or more of the selected exposure points. Because the grid is fixed and the moments are averaged over time, this is not an omnibus test over the entire support and nonlinear departures from the affine specification that offset across dates need not be detected. Technical conditions and the variance estimator are provided in Section [SA-II.10](#) of SA-Theory.

6 Empirical application

In this section, we introduce a novel risk factor and show that it predicts both the cross-sectional and time-series behavior of U.S. stock returns. In Section [SA-IV.2.2](#) of SA-Applied, we present a second empirical application that revisits the capital share factor of [Lettau et al. \(2019\)](#). Together, these applications illustrate the practical value of the new theoretical and inferential results developed earlier in the paper.

Our risk factor is a new measure of the business credit cycle. Standard measures of the business credit cycle use ratios of credit aggregates to measures of output. Although theoretically

appealing, these ratios can be difficult to interpret because their movements may reflect changes in the composition of the underlying aggregates rather than changes in credit conditions relative to economic activity. Here we take a different approach. We rely on the Federal Reserve’s Senior Loan Officer Opinion Survey²⁰ (SLOOS) to measure credit conditions and the ISM Manufacturing Index to measure economic activity. This construction has three distinct advantages. First, as the SLOOS and ISM can both be expressed as diffusion indices, their behavior is less sensitive to changes in the structure of the economy. Second, they are much more timely than credit aggregates and national accounts data which tend to be released with a substantial lag. Third, they are not subject to revision outside of updated seasonal factors for the ISM index. Thus, we have a timely factor which we can evaluate in real time with no look-ahead bias.

Our factor is constructed as

$$\text{BCF}_t = \left(\frac{1}{2} \cdot \text{SLOOS}_t + 50 \right) + \text{ISM}_t, \quad (6.1)$$

where SLOOS_t is the net percentage of large domestic banks tightening standards for commercial and industrial loans to all firms and ISM_t is the ISM Manufacturing Index. For both series we date each observation to the month in which it became publicly available.²¹ Although both the SLOOS and the ISM are based on questions with a similar format, they are scaled differently and so the affine transformation of the SLOOS is implemented so that they are both on the same scale (between 0 and 100). To understand why this is (the inverse of) a credit-to-output type measure, note that a fall in the SLOOS corresponds to easier lending standards and a fall in the ISM to weaker economic activity. Thus, a low value of the business-credit factor is intended to capture greater credit availability relative to economic activity. The reverse interpretation applies when the business-credit factor is high. Our factor series begins in January 1965 representing the first SLOOS observation.

As a preliminary check for the validity of our factor, we assess its ability to predict future market

²⁰For background information on the SLOOS and its relationship to lending and economic activity, see [Lown et al. \(2000\)](#) and [Lown and Morgan \(2006\)](#).

²¹The Senior Loan Officer Opinion Survey is currently conducted on a quarterly basis. To construct a monthly series we keep the SLOOS value constant until a new value is available. For the period from 1984m1 through 1990m1, the credit standards question was not included in the SLOOS, and we use the net willingness to make consumer installment loans by large domestic banks, with its sign reversed to match the tightening convention. For the initial period from 1965m1 through 1967m9, we use the credit-standards series for all banks and all firms.

returns. Motivated by equation (2.1), we ask whether the persistence of the factor translates into predictability of future equity returns. We estimate predictive regressions of the form

$$r_{t,t+\ell} = a^{(\ell)} + b^{(\ell)}z_t + v_{t+\ell}^{(\ell)}, \quad (6.2)$$

where $r_{t,t+\ell}$ is the cumulative market excess return over the following ℓ months and z_t denotes the predictor under consideration. We compare our risk factor with the benchmark predictors studied by [Welch and Goyal \(2008\)](#). In Table 1 we present in-sample R^2 from predictive regressions for forecast horizons of 1, 3, 6, and 12 months ahead. The first fourteen rows report results for the benchmark predictors. The row labeled “CGP” reports results using only the SLOOS component of our factor, as in [Chava et al. \(2015\)](#), while the row labeled “BCF” reports results for our business-credit factor. The results are stark. The in-sample R^2 from our new risk factor far outstrips that of the other predictors considered. The out-of-sample results, reported in Table SA-IV.2.1 of SA-Applied, lead to the same conclusion, namely, that our risk factor outperforms each of the other predictors by a wide margin.

We now investigate how the business-credit factor performs in explaining the cross-section of equity returns. We use monthly raw returns from the Center for Research in Security Prices (CRSP) from January 1965 through December 2019. We retain common shares listed on the New York Stock Exchange, American Stock Exchange, or Nasdaq and follow [Bali et al. \(2016\)](#) in incorporating delisting returns. The first 120 months are reserved for initial first-step estimation, so the reported expected-return estimates cover January 1975 through December 2019. We select the first-step window H_{it} and second-step number of portfolios J_t using the data-driven approach described in Section SA-III.1 of SA-Applied. Within each window, we use all available nonmissing observations for an asset and require at least 36 usable prior observations to estimate β_{it} and include the asset in that month’s portfolio sort. Further implementation details are provided in Section SA-III.4 of SA-Applied.

Figure 1 reports the estimated SACER, $\hat{\mu}(\beta)$, over a grid ranging from -1.25 to 1.25 in increments of 0.25 . The black line is the $(\mathbf{p}, \mathbf{q}) = (0, 0)$ point estimate. The estimated SACER exhibits a clear negative relationship with beta. It is around 1.5 percent per month at negative beta values and around 1 percent at positive beta values. The overall pattern indicates that greater exposure

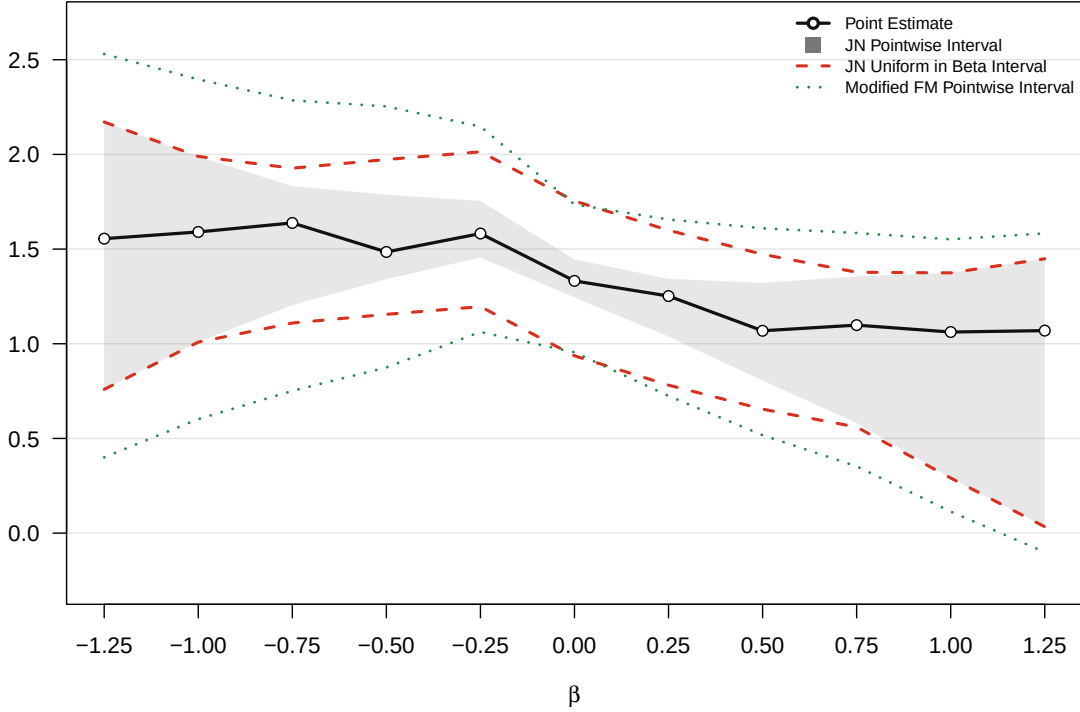
Table 1: **In-sample Predictive Regressions** This table reports R^2 (in percent) from predictive regressions of cumulative market excess returns on an individual predictor from [Welch and Goyal \(2008\)](#) for horizons of 1, 3, 6, and 12 months ahead. The row labeled “CGP” reports results for the SLOOS only portion of our risk factor as studied in [Chava et al. \(2015\)](#). The row labeled “BCF” reports results for our proposed risk factor. The sample period is 1965m1–2019m12.

	$\ell = 1$	$\ell = 3$	$\ell = 6$	$\ell = 12$
(log) Dividend Price Ratio	0.09	0.31	0.71	1.45
(log) Dividend Yield	0.11	0.32	0.74	1.48
(log) Earnings Price Ratio	0.03	0.04	0.06	0.25
(log) Dividend Payout Ratio	0.02	0.20	0.56	0.74
Stock Variance	1.01	0.11	0.15	0.65
Book-to-Market Ratio	0.00	0.01	0.07	0.12
Net Equity Expansion	0.15	0.22	0.48	1.25
Treasury Bill Yield	0.40	0.79	1.09	1.57
Long Term Treasury Yield	0.13	0.17	0.14	0.00
Long Term Treasury Return	1.16	0.71	1.88	1.34
Term Spread	0.52	1.38	2.50	6.99
Default Yield Spread	0.26	0.82	2.31	2.88
Default Return Spread	0.30	0.24	0.29	0.02
(lagged) Inflation	0.01	0.46	1.84	2.29
CGP	1.26	3.04	3.65	4.06
BCF	2.89	7.87	10.82	12.68

to the business-credit factor is associated with lower expected returns.

We use our new theoretical results to conduct inference on the SACER. In the figure, the grey region is the 95% pointwise interval based on our full variance estimator, $\widehat{\Sigma}(\beta)$ (denoted JN), while the green dotted lines delimit the 95% pointwise interval based on the modified FM variance estimator, $\widehat{\Sigma}_{\text{FM}}(\beta; 1)$. Both pointwise intervals use $(\mathbf{p}, \mathbf{q}) = (1, 1)$, and therefore need not be centered on the black line. Using our data-driven choices of tuning parameters, these intervals are asymptotically valid at fixed nonzero evaluation points (see Lemma [SA-III.1.1](#) of SA-Applied). The red dashed lines report the 95% JN uniform-in-beta interval discussed in Remark [4.7](#). In this application, the estimated boundaries are approximately $\beta = \pm 0.79$, so the uniform adjustment applies to the seven reported grid points from -0.75 through 0.75 . Consistent with the theory, Figure [1](#) shows that the uniform-in-beta adjustment widens the interval near zero, where the risk-factor component of the variance becomes small, while leaving the JN pointwise interval unchanged away from zero. The figure also shows that the modified FM pointwise intervals are wider than the JN pointwise intervals at every reported value of β .

Figure 1: **Inference on the SACER.** The black line reports the $(\mathbf{p}, \mathbf{q}) = (0, 0)$ estimate of the SACER. The grey region is the 95% JN pointwise interval. The red dashed lines are the 95% JN uniform-in-beta intervals, and the green dotted lines are the 95% modified FM pointwise intervals. All three intervals use $(\mathbf{p}, \mathbf{q}) = (1, 1)$. Results are reported for monthly returns (in percent). The sample period is 1965m1–2019m12 with a 120-month burn-in period.



We quantify the economic magnitude of the downward slope in Figure 1 using a high-minus-low contrast. Table 2 provides results for the contrast between $\beta = -0.75$ and 0.75 . We choose these two values as the contrast endpoints since they reside in the realized support in almost every month (both endpoints lie within the realized range in 94.6% of the months). The point estimate of this contrast, based on the $(\mathbf{p}, \mathbf{q}) = (0, 0)$ estimator, is -0.540 percentage points per month corresponding to the difference between the two points on the black line in Figure 1. Table 2 presents the results from our new inference procedures. The JN standard error is 0.304 , giving a (two-sided) p -value of 0.070 . The modified FM variance estimator produces a slightly smaller standard error, giving a two-sided p -value of 0.054 . Thus, both the JN and modified FM procedures reject a zero contrast at the 10% level. As a reference, using $\hat{\Sigma}_f$ alone gives a smaller standard error and a p -value of 0.019 . Although both endpoints are away from zero, so the idiosyncratic component is asymptotically of smaller order, we recommend the full JN variance estimator because it retains

this source of uncertainty. Finally, applying the affine specification test from Section 5.2 to the seven central grid points yields a Wald statistic of 7.871, with an associated chi-squared p -value of 0.344. Thus, we do not reject the null of an affine specification at conventional significance levels.

Table 2: **High Minus Low Contrast.** The table reports inference for $\hat{\mu}(0.75) - \hat{\mu}(-0.75)$ in monthly percentage points. Each row reports the standard error, 90% confidence interval, and two-sided normal-approximation p -value under the JN, modified FM, or $\hat{\Sigma}_f$ -only variance estimator. The confidence intervals are centered at the $(\mathbf{p}, \mathbf{q}) = (1, 1)$ estimate.

Variance estimator	SE	90% CI	p -value
JN	0.304	$[-1.049, -0.050]$	0.070
Modified FM	0.285	$[-1.019, -0.080]$	0.054
$\hat{\Sigma}_f$ only	0.234	$[-0.934, -0.165]$	0.019

7 Conclusion

Beta-sorted portfolios are a commonly used empirical tool in asset pricing. In a first step, time-varying factor exposures are estimated by rolling regressions of asset returns on observable risk factors to measure how returns co-vary with those factors. In a second step, assets with similar risk-factor exposures are grouped into portfolios, whose returns are compared across different exposure levels. Yet the simple and intuitively appealing algorithm belies a more complicated statistical setting involving a two-step procedure in which both stages involve nonparametric estimation.

We provide a comprehensive econometric framework that rationalizes this commonly used estimator and links its properties to the first-step estimation window and the number of portfolios formed in the second step. We establish uniform consistency and pointwise asymptotic normality for the time-averaged beta-sorted portfolio estimator, showing that its rate of convergence is $\sqrt{T}$ at fixed nonzero risk-factor exposures but a faster rate of $\sqrt{nT/J^d}$ at the origin. We also show that certain weighted linear combinations can attain this faster rate of convergence. We introduce a new variance estimator, characterize the properties of the ubiquitous Fama-MacBeth variance estimator, and provide inference procedures for hypotheses of interest. Finally, we identify limitations of current practice and provide guidance on implementation and interpretation.

We apply our new theoretical results in two substantive empirical applications. First, we introduce a new risk factor designed to capture the business-credit cycle. We find a downward-sloping relationship between expected returns and exposure to this factor, with a statistically significant

high-minus-low contrast using our proposed inference procedures. Second, we revisit the capital share risk factor proposed by [Letttau et al. \(2019\)](#). Using our new inference procedures, we do not find a statistically significant relation in the cross-section of U.S. equity returns.

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